

Exploiting a high-performance magnesium-fluoride battery prototype enabled by anion-receptor-mediated electrolyte

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 Check for updatesKeyi Chen^{1,2,3}, Meng Lei^{1,2,3}, Tengfei Wang^{1,3}, Guyue Li^{1,2,3} & Chilin Li^{1,2,3} ✉

Rechargeable magnesium batteries have emerged as a competitive next-generation energy storage technology owing to their abundant resources and inherent safety. The development of Mg batteries, especially regarding specific energy, has been compromised by the lack of high-performance positive electrodes. Here, we design an anion-receptor-mediated all-phenyl-complex electrolyte that enables the electrochemical cooperation between fluoride and magnesium, and develop a Mg || FeO_{0.7}F_{1.3} battery prototype. The anion receptor of tris(pentafluorophenyl)borane alters the chemical compositions of electrolyte and regulates its solvation structures via dipole-ion and dipole-dipole interactions with chloride-based ionic species and solvent molecules. The tailored electrolyte mitigates the chloride-originated anodic instability and upgrades the desolvation kinetics of cationic carriers. The dual-cation co-driven conversion-type Mg || FeO_{0.7}F_{1.3} batteries exhibit the high reversible capacities in a wide temperature range, delivering 354 mAh g⁻¹ at 25 °C and 177 mAh g⁻¹ at -20 °C. Moreover, the high reversibility over 500 cycles with a low capacity decay rate of 0.054% per cycle can be realized when confined to intercalation chemistry. This work provides an anion receptor-based electrolyte design strategy that expands positive material horizons for magnesium batteries to high-energy fluorides.

Although lithium-ion batteries (LIBs) have dominated the energy storage market, including portable electronic devices and electric vehicles, the inherent problems of high cost (resource shortage) and safety concern (Li dendrite growth) limit their applications in large-scale energy storage systems¹. As an alternative, magnesium metal batteries (MMBs) have received increasing attention owing to the abundant Mg element (the 8th most abundant in the earth's crust) and the utilization of Mg metal negative electrode, which can offer the high volumetric capacity of 3833 mAh cm⁻³ (~1.8 times that of Li metal) and relatively low redox potential of -2.37 V (versus standard hydrogen electrode)². In particular, due to the less reactive nature compared with Li metal,

Mg metal shows a low tendency to the occurrence of dendrite propagation and internal short circuit of MMBs during electrochemical stripping/plating cycling². Therefore, MMBs have great potential to develop into the next-generation energy storage technology (beyond LIBs) that can meet the requirements of safety, sustainability and high performance. However, the bivalent Mg²⁺ with high charge density induces the great electrostatic interaction with host lattices and strong ligand binding in solvation, which results in sluggish mass transport kinetics in bulk electrode and electrolyte as well as at their solid-liquid interfaces^{3,4}. While substantial progress has been made in the electrode interface engineering that facilitates the reversible Mg stripping/

¹State Key Laboratory of High Performance Ceramics and Superfine Microstructure, Shanghai Institute of Ceramics, Chinese Academy of Sciences, Shanghai, China. ²Center of Materials Science and Optoelectronics Engineering, University of Chinese Academy of Sciences, Beijing, China. ³CAS Key Laboratory of Materials for Energy Conversion, Shanghai Institute of Ceramics, Chinese Academy of Sciences, Shanghai, China.

✉ e-mail: chilinli@mail.sic.ac.cn

plating^{5–7}, the further advancement of MMBs is still hindered by lacking high-energy and high-capacity positive electrodes.

Up to now, the widely used positive electrode materials in MMBs are mainly limited to chalcogenides and oxides. Transition-metal chalcogenides with “soft” anion sublattices can weaken the Mg^{2+} -host interaction due to the enhanced structure polarizability, thereby contributing to the bivalent cation mobility in solid-state frameworks^{8,9}. But even so, the reversible Mg-intercalated capacities are typically less than 110 mAh g^{-1} for Chevrel Mo_6S_8 and 200 mAh g^{-1} for thiospinel Ti_2S_4 ^{10–12}, even under the favorable operating conditions (e.g., a low current rate of 0.1 C and an elevated temperature of 60°C). Although the conversion-type sulfides (e.g., FeS_2 and CuS_2) can deliver the higher specific capacities, the cycling stability is severely compromised by poor reaction kinetics and potential shuttle effect of polysulfide intermediates^{13,14}. More unfortunately, chalcogenide compounds typically only offer a low discharge potential of -1.0 V vs. Mg^{2+}/Mg , which inherently restricts the specific energy improvement for MMBs. Substituting chalcogenide with oxide clusters (e.g., V_2O_5 and MnO_2) can theoretically raise the output voltage^{15,16}. However, Mg^{2+} ions suffer from a high diffusion barrier in oxide lattices due to their strong affinity to O^{2-} , which is also prone to induce the extrusion of Mg oxide rather than following a complete intercalation route during electrochemical reaction¹⁶. To optimize the Mg^{2+} -diffusion environment, structure pillaring is an effective strategy, especially for two-dimensional layered materials with high structure tunability. The bilayer-structured $\text{V}_2\text{O}_5\cdot\text{H}_2\text{O}$ materials have been developed for hosting multivalent ions, benefiting from the dual effects of interlayer expansion and electrostatic shielding by lattice water^{17–19}. For instance, the hydrated V_2O_5 with Mg^{2+} ion pre-intercalation enables a reversible and stable Mg^{2+} -storage capacity of $\sim 164 \text{ mAh g}^{-1}$ under a specific current of 0.1 A g^{-1} . Despite the enhanced Mg-insertion reactivity, it is at the expense of practical output voltage (in the range of -1.0 – -1.0 V) since using active carbon as the negative material instead of Mg metal. The compromised choice of negative electrode should be attributed to the electrolyte incompatibility and inevitable release of crystal water into electrolyte, which induce the undesired interfacial side reactions and surface passivation of Mg metal^{17,18}. In addition to H_2O , some organic molecules such as poly(3,4-ethylenedioxythiophene) (PEDOT) and cetyltrimethylammonium bromide (CTAB) also can serve as intercalation agents to upgrade the Mg^{2+} diffusion kinetic^{14,20}. Nevertheless, introducing the electrochemically inactive and large-size pillars would markedly decrease the specific energy in both gravimetric and volumetric levels (e.g., the interlayer spacing of V_2O_5 is more than doubled by PEDOT pillaring). Overall, it is necessary and urgent to explore the energy-dense positive electrode materials for advanced MMBs.

Compared with oxides and chalcogenide counterparts, metal fluorides with the high electronegativity F-ligand can yield the higher thermodynamic equilibrium potentials²¹. Among them, the resource-abundant and cost-efficient iron trifluoride (FeF_3) has been widely investigated recently and identified as promising energy-dense positive electrode (with a considerable value of 1947 Wh kg^{-1}) in LIBs, based on a thermodynamic potential of -2.73 V (vs. Li^+/Li) and a high theoretical capacity of 712 mAh g^{-1} via three-electron transfer reaction²². Therefore, exploring the electrochemical cooperation between iron fluoride positive electrode and Mg metal negative electrode will promote the development of high-energy Mg-based battery prototypes. However, conversion-type magnesium-fluoride batteries may be challenging in efficient operation when relying exclusively on Mg^{2+} ion shuttling. This is primarily due to the slow kinetics associated with divalent carriers, which involve the solid diffusion through dense and rigid fluoride structure, as well as the breaking and reformation of strong metal-fluorine bonds²¹. Inspired by Mg-based hybrid batteries, magnesium-fluoride conversion systems are expected to be activated by using dual-salt electrolytes, which enable the combination of

kinetically favorable monovalent (e.g., Li^+) chemistry at positive electrode and thermodynamically dominant Mg stripping/plating at negative electrode²³. In this regard, fluoride choice for Mg-based hybrid batteries can more or less follow the structure design method that has been established in lithium batteries. Given the above-mentioned negative impacts of lattice water on electrolyte stability and compatibility with Mg metal, the typical open-framework strategy via H_2O stuffing seems unavailable on conversion-type fluorides involving phase-separation reactions in MMBs. Mixed anion chemistry with O-substitution for fluorides could improve the intrinsic electronic conductivity and ionic mobility due to the weakened bond ionicity²⁴. Moreover, it is also expected to regulate the phase evolution pathway and mitigate the phase separation within a confined voltage range²⁵. Consequently, the kinetic capacity can be enhanced despite a certain loss in thermodynamic capacity.

Herein, we developed a rechargeable magnesium-fluoride battery prototype that promises a further improvement in the specific energy of MMBs. The electrochemical system comprises the iron oxyfluoride ($\text{FeO}_{0.7}\text{F}_{1.3}$) positive electrode and the electrolyte based on an all-phenyl-complex (APC) solution mediated by an anion receptor of tris(pentafluorophenyl)borane (TPFPB). As a strong Lewis acid, the TFPB molecule exhibits the strong interactions with Cl-based ionic species and alters their charge distribution through electron-withdrawing effects. Meanwhile, it affects the equilibrium species of Lewis acid-base reactions in APC-type solution with tetrahydrofuran (THF) solvent, resulting in the decreased concentration of magnesium dimer cation $[\text{Mg}_2\text{Cl}_3\cdot 6\text{THF}]^+$ (a major culprit for corrosion). Consequently, the modified electrolyte is able to inhibit the anodic corrosion on current collector and enhance the high voltage tolerance (up to 3.71 V vs. Mg^{2+}/Mg using Mo foil). Moreover, the TFPB-induced dipole-dipole/ion interactions loosen the solvation structures of APC-type electrolyte, thereby reducing the desolvation barriers for both Mg^{2+} and Li^+ cations. Benefitting from the enhanced oxidative stability and interfacial reaction kinetics of electrolyte, the conversion-type $\text{Mg}|\text{FeO}_{0.7}\text{F}_{1.3}$ batteries deliver the reversible capacities as high as 354 mAh g^{-1} at room temperature (25°C) as well as 177 mAh g^{-1} at low temperature (-20°C). When confining the fluoride activation to intercalation reaction stage, the cells enable a high average discharge voltage of 1.77 V and good cycling stability, sustaining over 500 cycles with a low capacity decay rate of 0.054% per cycle. This work broadens the positive electrode material option for MMBs and provides a promising avenue to develop an energy-dense, high-safety, and low-cost energy storage technology. The proposed anion receptor chemistry offers an effective solution to the interfacial issue of chloride-based electrolytes in high-voltage applications.

Results

Physical and electrochemical properties of APC-type solutions

As a proven magnesium electrolyte, the APC-type solution is commonly characterized by low cost, high ionic conductivity, and high compatibility with Mg metal²⁶. However, the presence of Cl-containing components (e.g., $[\text{Mg}_2\text{Cl}_3\cdot(\text{THF})_6]^+$) would cause the severe anodic corrosion to non-noble metal current collectors (such as Al and stainless steel, denoted as SS) at the potentials exceeding -2.0 V , which limits the choice of high-voltage positive electrodes and the improvement of specific energy for MMBs^{27,28}. To tackle this issue, in this context, the anion acceptor TFPB is introduced into the APC solution to regulate Lewis acid–base reactions and solvation structure. Specifically, a typical hybrid solution containing 0.25 M APC and 1.0 M LiCl is used as the baseline electrolyte (denoted as LiCl-APC). The anion-receptor-mediated electrolyte is prepared by adding 0.02 M TFPB into the baseline LiCl-APC solution, and it is denoted as LiCl-APC-TPFPB.

Quantum chemical calculations based on density functional theory (DFT) were performed to quantitatively evaluate the interactions

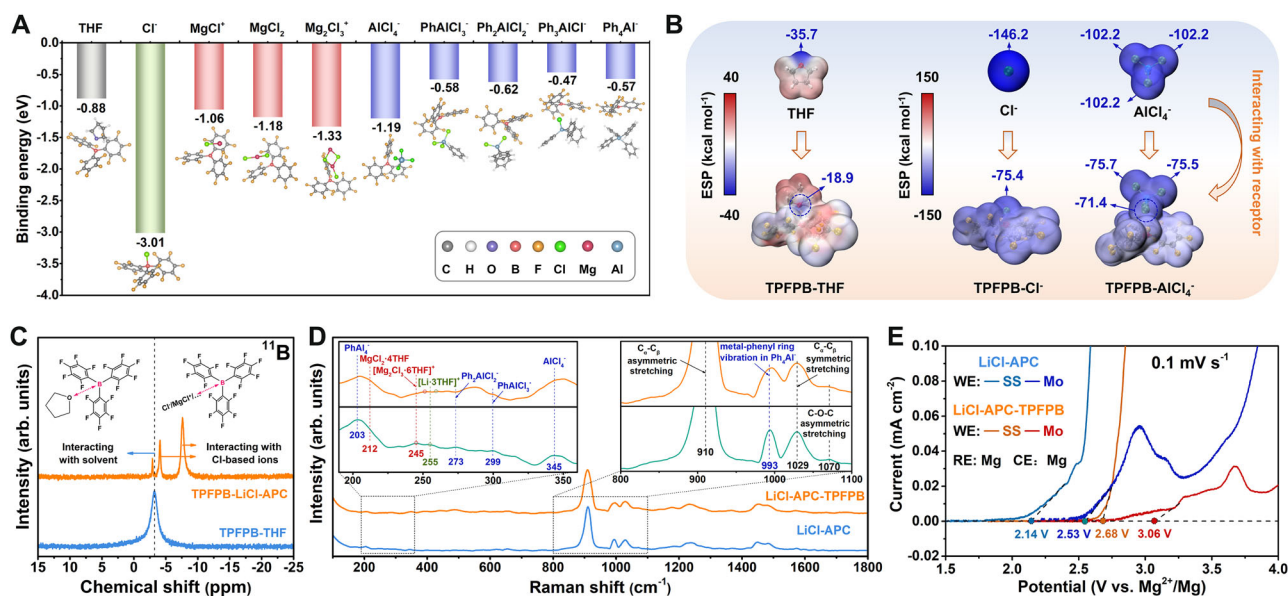


Fig. 1 | Electrolyte chemistry of APC-type solution regulated by anion receptor TFPFB. **A** Binding energies of TFPFB with individual components in LiCl-APC-based electrolyte and corresponding structure configurations after adsorption optimization by DFT calculation. **B** Calculated electrostatic potential maps (ESP) of THF solvent, Cl⁻ and AlCl₄⁻ anions without and with TFPFB coordination based on DFT calculation. **C** ¹¹B NMR spectra of TFPFB-LiCl-APC (top) and TFPFB-

THF (bottom) solutions. **D** Raman spectra of LiCl-APC-TPFPB (top) and LiCl-APC (bottom) electrolytes. Insets: magnified spectra in different Raman shift regions. **E** LSV curves of LiCl-APC solutions without and with TFPFB additive in a three-electrode configuration using Mg foils as reference electrode (RE) and counter electrode (CE), and stainless steel (SS) or Mo foil as working electrode (WE), respectively. The LSV tests were performed at 25 °C.

between anion receptor and solvent and Cl-based species in LiCl-APC electrolyte. As shown in Fig. 1A, the TFPFB molecule with electron-deficient boron center exhibits an attraction to the THF solvent with electron-rich ethereal O group, resulting in a binding energy of -0.88 eV. In comparison, the electrostatic attraction with chlorine ligand is stronger, resulting in the more negative binding energies for Cl⁻ ion (-3.01 eV), AlCl₄⁻ (-1.19 eV), and magnesium chloride complexes (-1.06, -1.18, and -1.33 eV for MgCl⁺, MgCl₂, and Mg₂Cl₃⁺, respectively). Note that the absorbed Mg₂Cl₃⁺ cluster on TFPFB undergoes structural deformation (Supplementary Fig. 1), where one of the chlorine atoms breaks the original Mg-Cl-Mg bridge and approaches the boron center of anion receptor. It seems that the introduction of TFPFB is unfavorable for the stable existence of Mg₂Cl₃⁺ species in APC solution. Due to the steric hindrance effect between TFPFB and phenyl groups, TFPFB shows the weak attractions to phenylaluminum chloride Ph_nAlCl_{4-n} (*n* = 1–4) complexes with the low binding energies ranging from -0.47 to -0.62 eV. Based on the optimized structure configurations, surface electrostatic potential (ESP) analysis was performed to visualize and quantify the variation in charge distribution of anion/solvent without and with TFPFB coordination. As illustrated in Fig. 1B, the negative charge (blue area) is mainly localized at the oxygen atom of THF solvent and the chlorine atoms of Cl⁻/AlCl₄⁻ anions. When these solvents and anions are absorbed onto TFPFB with the electron-withdrawing character, the negative charge around the ethereal O group and Cl ligand is markedly dispersed, as indicated by the diluted color and less negative ESP values in the associated blue regions. The less negative ESP value means the weaker binding to Li⁺/Mg²⁺ cation and the decreased solvation energy²⁹. Therefore, it implies the favorable desolvation and dechlorination processes for the TFPFB-mediated solvation environment. On the other hand, the decreased surface electron density of THF solvent and Cl-based anions is expected to improve the oxidation stability and alleviate the anodic corrosion in APC electrolyte.

To further identify the interaction behavior between TFPFB and the components in LiCl-APC electrolyte, nuclear magnetic resonance (NMR) spectroscopy and Raman spectroscopy were carried out. In the

¹¹B NMR spectra (Fig. 1C), a prominent peak associated with the polar solute-solvent interaction is found in the upfield region with a chemical shift (δ) of -3.22 parts per million (ppm) for the case of TFPFB-THF solution³⁰. The addition of chloride salt (for LiCl-APC-TPFPB solution) results in two pronounced peaks newly emerging in the upfield region with the larger δ values of -4.11 and -7.62 ppm, respectively, compared with that of ¹¹B signal for TFPFB-THF interaction. These two peaks should arise from the ion-dipole interaction between TFPFB and Cl-based ions, leading to an enhanced shielding effect around the boron atom as the electron-rich Cl-ligand approaches. Note that the TFPFB-THF interaction is weakened due to the less competitive coordination with THF than Cl-ligand, as reflected by the attenuated signal with a downfield shift of 0.32 ppm (to -2.90 ppm). As a strong Lewis acid, TFPFB will affect the equilibrium species of reactions between phenylmagnesium chloride (PhMgCl) as a Lewis base and AlCl₃ as a Lewis acid, while also regulating the solvation structure in LiCl-APC electrolyte. Raman spectroscopy was performed to identify the TFPFB-mediated electrolyte chemistry, as shown in Fig. 1D. For the electrolytes with and without TFPFB, both the Raman spectra are dominated by the multiple vibration modes generated from THF solvent, such as the C_α-C_β asymmetric stretching with a prominent peak at 910 cm⁻¹ and the C_α-C_β symmetric stretching with a shoulder peak at 1029 cm⁻¹³¹. Compared with the additive-free case, the TFPFB addition causes the broadening of C-O-C stretching vibration peak at 1070 cm⁻¹, likely reflecting the changed coordination environment of THF by receptor. Driven by the TFPFB-induced interaction, the Raman peak related to [Mg₂Cl₃·6THF]⁺ becomes somewhat blurred and shows an evident blue shift from 245 cm⁻¹ to 252 cm⁻¹. Meanwhile, the Raman signal related to MgCl₂·4THF at 212 cm⁻¹ is intensified. It is deduced that TFPFB appears to inhibit the reaction equilibrium in which MgCl⁺ and MgCl₂ together form the Mg₂Cl₃⁺ species. The reduction in Mg₂Cl₃⁺ is consistent with the result of its structural instability when interacting with TFPFB, as indicated by the above DFT calculation (Supplementary Fig. 1). As a major culprit for corrosion, the decreased concentration of [Mg₂Cl₃·6THF]⁺ is expected to alleviate the corrosivity of TFPFB-mediated electrolyte²⁸. Note that MgCl⁺ is considered

to be undetectable in Raman spectroscopy³². The Raman signals associated with Al-based ionic species have a certain degree of blue shift, also resulting from the coordination effect induced by the Lewis acidic receptor^{32,33}. It can be observed that the peak at 993 cm^{-1} attributed to the metal-phenyl ring vibration of Ph_4Al^- decreases in intensity, while the peak at 345 cm^{-1} attributed to AlCl_4^- increases in intensity. The decreased concentration of the most oxidation-vulnerable Ph_4Al^- in exchange for the increased concentration of the most oxidation-resistant AlCl_4^- can contribute to the enhancement of high-voltage tolerance for the TFPB-containing APC system.

To confirm the oxidation stability of APC-based electrolytes towards different metallic current collectors, linear sweep voltammetry (LSV) measurements were conducted using the $\text{Mg}|\text{SS}$ or $\text{Mg}|\text{Mo}$ asymmetric cells at a scan rate of 0.1 mV s^{-1} (Supplementary Fig. 2). In the cases of SS current collectors, LiCl-APC electrolyte shows a low onset potential of -2.18 V with a subsequent booming anodic current response to corrosion. The addition of TFPB into the electrolyte enables a higher endurable potential of -2.90 V vs. Mg^{2+}/Mg . The optical images of SS current collectors after LSV measurements are presented in Supplementary Fig. 3. After the anodic scanning to 3.3 V in the baseline APC-type solution, severe corrosion occurs on the SS surface, which is fully covered by thick and white by-products. For the TFPB-containing system, even at a higher termination potential of 4.0 V , the anodic current is decreased and only some thin coverages are observed in the local region, indicating that the corrosion phenomenon is significantly mitigated. As expected, the anti-corrosion ability could be enhanced when adopting Mo foil as the current collector instead of SS. Even then, the onset potential is only slightly raised to -2.46 V for the TFPB-free system. The LiCl-APC-TFPB electrolyte performs a highly stable potential up to -3.71 V before a rapid increase of current response. To avoid overestimating the electrochemical window of electrolyte due to overpotential on the counter electrode, we re-evaluated the intrinsic oxidative stability of two APC-type electrolytes in a three-electrode configuration, using Mg foils as reference and counter electrodes (as shown in Fig. 1E). In the baseline LiCl-APC electrolyte system with SS and Mo working electrodes, the oxidation onset potentials of -2.14 and -2.53 V vs. Mg^{2+}/Mg are observed, respectively. When the electrolyte system contains TFPB, these endurable potentials are increased to -2.68 V for SS and -3.06 V for Mo, accompanied by a decrease in the anodic current response. Overall, the extended electrochemical stability window and the alleviated anodic corrosion to non-noble metal current collectors give support to the utilization of APC-based electrolytes.

As shown in Supplementary Fig. 4, the ionic conductivities of baseline APC-type electrolyte and the TFPB-containing electrolyte were measured at 25°C . The addition of TFPB additive, which has a relatively large molecular structure, would more or less increase the viscosity of the electrolyte and decrease its bulk ion mobility. On the other hand, as an anion receptor, TFPB can enhance the dissociation of chloride salts by coordinating with Cl ligands, thereby promoting the cation migration in bulk electrolyte and across interfaces/interfaces. As a result, two-type electrolytes exhibit the comparable ionic conductivities of 2.96×10^{-3} and $2.92 \times 10^{-3}\text{ S cm}^{-1}$ for LiCl-APC and LiCl-APC-TFPB, respectively. To evaluate the reversibility of Mg plating/stripping in the baseline APC-type electrolyte and TFPB-containing electrolyte, we galvanostatically measured the coulombic efficiency (CE) of $\text{Mg}|\text{Cu}$ asymmetric cells at a current density of 0.5 mA cm^{-2} and an areal capacity of 1.0 mAh cm^{-2} (Supplementary Fig. 5). As shown in Supplementary Fig. 5A, the initial CE of $\text{Mg}|\text{Cu}$ asymmetric cell in LiCl-APC electrolyte is 91.89% and the CE gradually increases in subsequent cycles, reaching 98.68% and 99.37% after 10 and 50 cycles, respectively. However, an abnormal increase in overpotential, accompanied by a jitter in charge curve, is observed before the Mg stripping step is completed (Supplementary Fig. 5B). This unstable Mg stripping behavior is likely due to the anodic corrosion of

Cu collector caused by the baseline APC-type electrolyte. For the asymmetric cell using LiCl-APC-TFPB electrolyte, the TFPB with electron-withdrawing character tends to undergo reduction decomposition and promote the formation of solid electrolyte interphase (SEI) during the first Mg plating step, resulting in a relatively low initial CE of 74.28%. Nevertheless, the CEs quickly recover to the values comparable to those of LiCl-APC case with the progress of cycling, reaching 95.50% and 99.26% after 10 and 50 cycles, respectively. Different from the control case, the $\text{Mg}|\text{Cu}$ cell with TFPB-containing electrolyte can maintain flat voltage profiles during long-term Mg plating/stripping cycles (Supplementary Fig. 5C), which should benefit from the key role of anion receptor in inhibiting Cl-associated corrosion. These results indicate the highly reversible and stable Mg plating/stripping behavior enabled by the TFPB-mediated electrolyte.

Solvation chemistry of APC-type electrolytes

To investigate the modulating effect of TFPB additive on the solvation chemistry of APC-type electrolyte, DFT calculations combined with NMR spectroscopic analysis were conducted (Fig. 2). In the APC-type solutions where Mg^{2+} and Li^+ ions are typically coordinated by a combination of Cl^- and THF ligands³², we employed quantum chemistry simulations to optimize the solvated configurations of both Mg-ion (i.e., $\text{Mg}^{2+}(\text{THF})_n\text{Cl}^-$) and Li-ion (i.e., $\text{Li}^+(\text{THF})_n\text{Cl}^-$), systematically comparing the cases with and without TFPB mediation (Supplementary Figs. 6–9). Although the receptor additive is not directly involved in the inner solvation sheath of cationic species, it can loosen the solvation structures via the TFPB-induced dipole-ion ($\text{B}^-\cdots\text{Cl}^-$) interaction with chloride ions and the dipole–dipole interaction ($\text{F}^-\cdots\text{H}-\text{C}$) with THF molecules (Fig. 2A). As shown in Fig. 2B, the introduction of TFPB extends the $\text{Mg}\cdots\text{Cl}$ distance between Mg^{2+} and Cl^- as well as the average $\text{Mg}\cdots\text{O}$ distance between Mg^{2+} and THF for Mg solvation complexes. Similar trends are seen for the Li solvation complexes in terms of $\text{Li}\cdots\text{Cl}$ and $\text{Li}\cdots\text{O}$ coordination distances (Fig. 2C). Note that TFPB exhibits a more significant weakening effect on the solvation structures of those lower-coordination-number complexes due to the reduced steric hindrance. To further understand the weak solvation effect, ^7Li NMR measurements were performed on the LiCl-APC and LiCl-APC-TFPB electrolytes (Fig. 2D). A downfield shift in the ^7Li signal of solvated lithium salt is observed upon the addition of TFPB to the baseline APC-type electrolyte, correspondingly moving from $\delta = 0.53$ to 0.69 ppm . This result confirms the deshielding effect of the receptor additive on the cations in electrolyte, which weakens the coordination of cations with electron-donating Cl^- ions and ethereal O groups of THF solvent.

The weakly solvating electrolytes usually have good desolvation capability³⁴. Here, we further evaluate the ion desolvation processes of solvation structures in the APC-type electrolytes, focusing on both the cation-solvent and cation-anion interactions. According to DFT calculations (Fig. 2B, C), the hexa-coordinated $\text{Mg}^{2+}(\text{THF})_5\text{Cl}^-$ and tetra-coordinated $\text{Li}^+(\text{THF})_3\text{Cl}^-$ are the thermodynamically stable solvation structures due to their lowest formation energies among various solvated Mg and Li configurations, respectively. Figure 2E, F compares the calculated desolvation energy profiles of these solvation structures with and without TFPB mediation. These energy profiles follow the typical desolvation path involving sequential solvent detachment and anion dissociation. For the bare $\text{Mg}^{2+}(\text{THF})_5\text{Cl}^-$ complex (Fig. 2E), the calculated energy barriers for stepwise stripping of THF molecules range from 0.98 to 1.46 eV , and they are much lower than 3.07 eV required for the subsequent stripping of Cl^- ion to yield a naked Mg^{2+} ion. This indicates that dechlorination is the rate-determining step for the whole Mg desolvation process in APC-type electrolyte. Regarding the anion receptor-mediated solvated complex, the energy barriers associated with stepwise desolvation processes are decreased via the interactions between TFPB and both Cl^- and THF. Notably, the energy barrier for rate-determining $\text{Mg}^{2+}\cdots\text{Cl}^-$ scission is substantially lowered

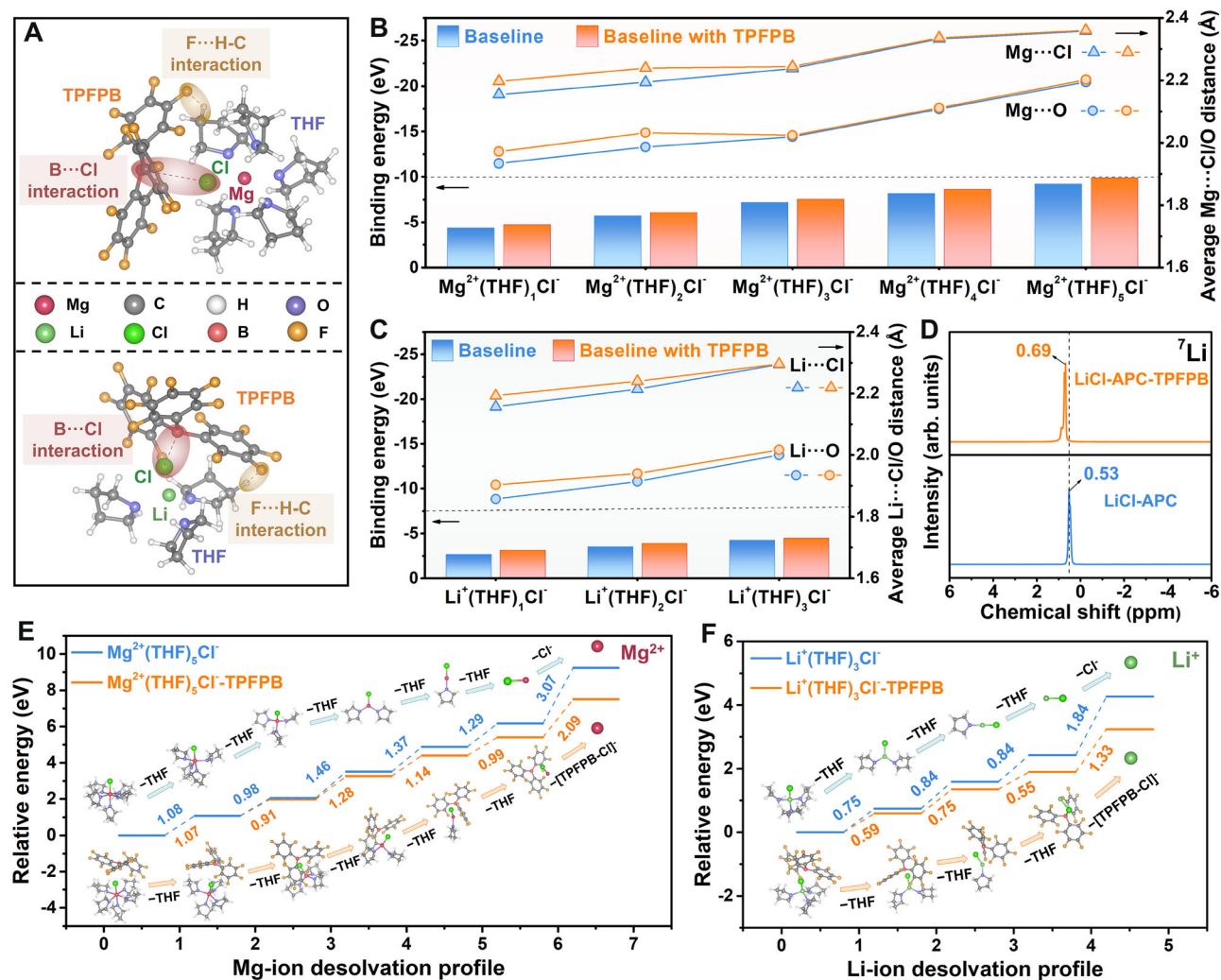


Fig. 2 | Solvation modulation for Mg-ion and Li-ion in APC-type electrolyte enabled by anion receptor TFPB. **A** Representative solvation structures for Mg-ion (top) and Li-ion (bottom) obtained by quantum chemistry simulations. Calculated binding energies for **B** Mg²⁺(THF)_nCl⁻ (n=1–5) and **C** Li⁺(THF)_nCl⁻ (n=1–3) solvation complexes without (baseline) and with TFPB. Corresponding average

Mg-Cl/O between Mg²⁺ and Cl⁻/THF and average Li-Cl/O distances between Li⁺ and Cl⁻/THF are counted. **D** ⁷Li NMR spectra of LiCl-APC-TFPB (top) and LiCl-APC (bottom) electrolytes. Comparison of desolvation energy profiles for **E** Mg-ion solvation cluster (Mg²⁺(THF)₅Cl⁻) and **F** Li-ion cluster (Li⁺(THF)₃Cl⁻) without and with TFPB mediation.

to 2.09 eV, and it is facilitated by the formation of thermodynamically stable [TFPB-Cl]⁻ complex. Due to the less concentrated charge for Li⁺ compared to Mg²⁺, the Li⁺(THF)₃Cl⁻ complex undergoes the more energetically favorable desolvation processes (Fig. 2F). Further, TFPB also plays an active role in the stripping of Li-solvation sheath, effectively reducing the corresponding energy barriers for four desolvation steps from 0.75, 0.84, 0.84, and 1.84 eV to 0.59, 0.75, 0.55, and 1.33 eV, respectively. In general, incorporating TFPB into the APC-type electrolyte enables the weakly solvated structures and kinetically promotes the desolvation of dual carrier ions (Li⁺ and Mg²⁺) based on a loose coordination chemistry. The positive effect is expected to accelerate the charge transfer reactions at electrode–electrolyte interface and improve the electrochemical behaviors of Mg-based batteries.

Electrochemical performance of Mg||FeO_{0.7}F_{1.3} batteries

To evaluate the electrochemical performance of the magnesium-fluoride prototype system, galvanostatic charge–discharge measurement was carried out on the Mg||FeO_{0.7}F_{1.3} batteries with two types of hybrid APC-based electrolytes. As shown in Supplementary Fig. 10, the FeO_{0.7}F_{1.3} positive electrodes can be electrochemically activated in Mg-

based battery systems under a voltage range of 0.5–2.5 V. The initial discharge reaction begins at -2.15 V with a prominent plateau at -1.65 V, corresponding to the Li-dominated intercalation in FeO_{0.7}F_{1.3} with Fe³⁺/Fe²⁺ reduction. Note that a short slant discharge curve above the main plateau should be attributed to the intercalation reaction in the minor FeF₃ phase in FeO_{0.7}F_{1.3}²⁵. The Li-intercalated (oxy)fluorides undergo Li⁺/Mg²⁺ co-driven conversion reactions within the voltage zone from -1.0 V to 0.5 V. Then, the conversion products (e.g., Fe, LiF and MgF₂) are sequentially oxidized into Fe²⁺/Fe³⁺-containing rutile/rocksalt phases through delithiation and demagnesiumation reactions during the subsequent charge process. The phase transformation mechanism will be verified and discussed in the later sections.

In both hybrid electrolyte systems, the Mg||FeO_{0.7}F_{1.3} cells could deliver a high initial discharge capacity over 450 mAh g⁻¹ at 100 mA g⁻¹ via multi-electron transfer reaction. However, there is a large irreversible capacity during the charge process (i.e., overcharge behavior) in the LiCl-APC system (Supplementary Fig. 10A), which should result from the severe interfacial side reactions due to the poor oxidation stability of the electrolyte. Moreover, the converted metallic iron in the cycled fluoride positive electrode is susceptible to anodic corrosion from the Cl-based electrolyte and has a potential catalytic effect on

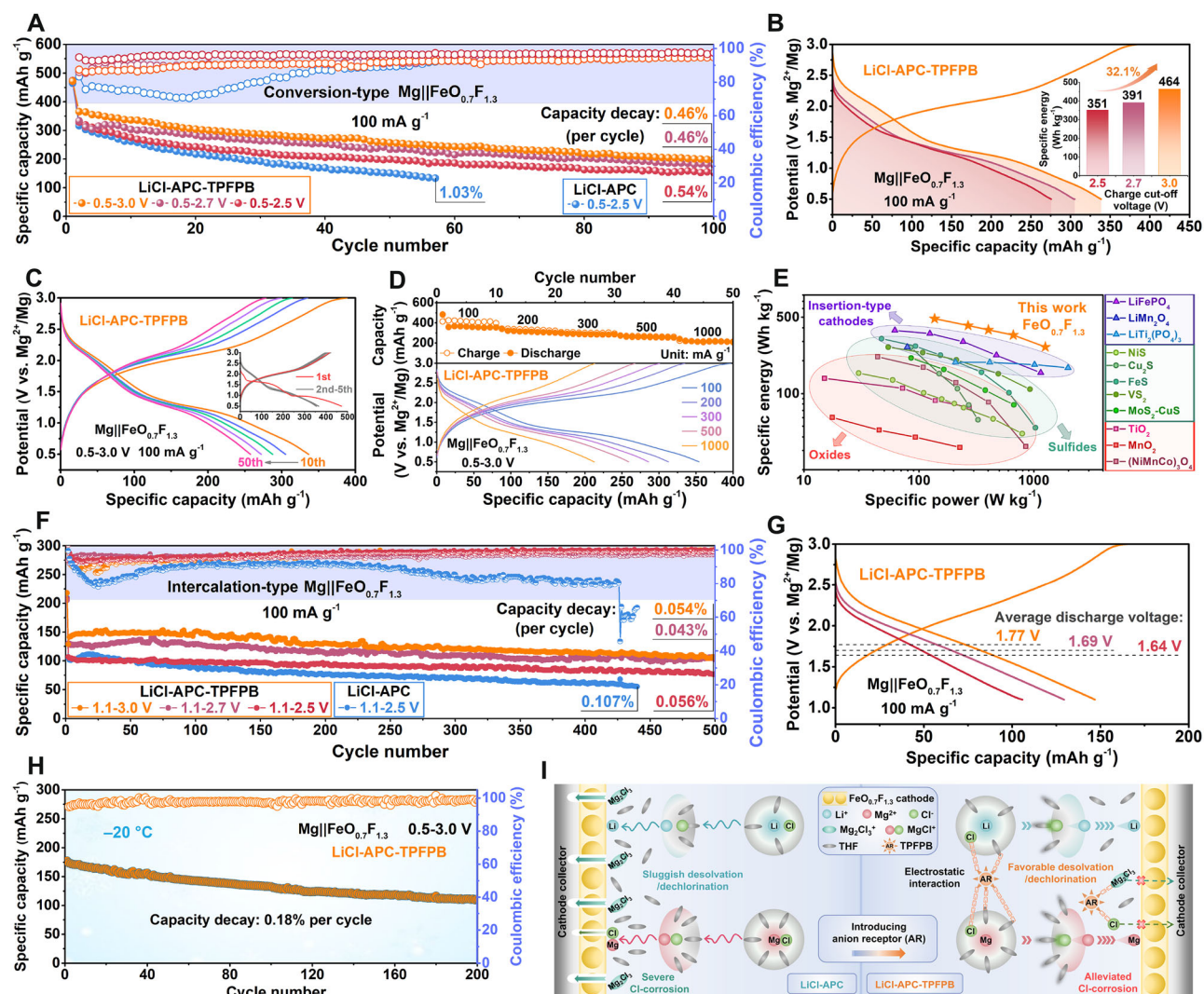


Fig. 3 | Electrochemical performance of $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells. A Discharge capacity and coulombic efficiency of conversion-type $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells based on LiCl-APC-TPFPB and LiCl-APC electrolytes at 100 mA g^{-1} . **B** Typical galvanostatic discharge curves and corresponding specific energies (inset) of TPFPB-containing systems with different charge cut-off voltages of 2.5 V, 2.7 V, and 3.0 V. **C** Discharge-charge profiles of $\text{FeO}_{0.7}\text{F}_{1.3}$ electrode with LiCl-APC-TPFPB electrolyte at different cycling stages in a voltage range of 0.5–3.0 V at 100 mA g^{-1} . **D** Rate performance of TPFPB-containing system depending on various specific currents from 100 to 1000 mA g^{-1} (upper) and corresponding charge-discharge curves (lower). **E** Ragone plots showing specific energies and powers of our $\text{FeO}_{0.7}\text{F}_{1.3}$ compared to representative

oxides, sulfides, and insertion-type positive electrodes for Mg-Li hybrid systems. Both specific energies and powers are calculated on the basis of weight of active fluoride materials. **F** Cycling performance of intercalation-type $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells based on LiCl-APC-TPFPB and LiCl-APC electrolytes at 100 mA g^{-1} . **G** Typical galvanostatic discharge curves and corresponding average voltages of TPFPB-containing systems with different charge cut-off voltages of 2.5 V, 2.7 V, and 3.0 V. **H** Cycling performance of TPFPB-containing system at 20 mA g^{-1} and a low temperature of -20°C . **I** Schematic illustration of TPFPB's role in regulating solvation structure and interfacial activity of APC-type electrolyte. All galvanostatic discharge-charge tests were performed at 25°C , except in the case of Fig. 3H.

interfacial side reactions³⁵. With the progress of cycling, this undesired interfacial behavior is gradually aggravated, along with a sharp decrease in CE (even as low as ~70%) and capacity (below 200 mAh g^{-1} after only 30 cycles), as shown in Fig. 3A. By comparison, using the LiCl-APC-TPFPB electrolyte with enhanced antioxidation ability, the $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cell performs the highly reversible charge-discharge reaction with well-defined voltage plateaus (Supplementary Fig. 10B). It enables the higher CE values of ~97% and improved cycling stability with a smaller average capacity decay rate of 0.54% per cycle after the initial activation cycle (vs. 1.03% for the baseline system, Fig. 3A). Given the proven high-voltage tolerance of the optimized electrolyte, we further raise the upper charge cut-off voltage to motivate the deeper redox reaction of fluoride in TPFPB-containing battery system. As shown in Fig. 3B, when the charge cut-off voltages are raised from 2.5 to 2.7 V and 3.0 V, the $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells display the upshifted discharge plateaus and increased reversible capacities from 276 to

304 mAh g^{-1} and 338 mAh g^{-1} , leading to the improved specific energies from 351 to 391 Wh kg^{-1} and 464 Wh kg^{-1} (increased by 32.1%, based on the mass of the active fluoride materials, inset of Fig. 3B), respectively. This positive effect is in agreement with the intensification of a couple of cathodic peaks (around 2.05 and 1.20 V) in cyclic voltammetry (CV) curves (Supplementary Fig. 11). The increase of charge cut-off voltage does not seriously compromise the interfacial stability for the 2.7 V class cells, which maintain the reversible charge-discharge profiles (Supplementary Fig. 12A) and high CE values approaching ~96% with cycling at 100 mA g^{-1} (Fig. 3A). Fluoride positive electrode enables a higher oxidation state at the raised upper cut-off voltage, which contributes to the prominent two-stage discharge plateaus for the 3.0 V class $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cell (Fig. 3C). Benefitting from the favorable $\text{Li}^+/\text{Mg}^{2+}$ desolvation and alleviated anodic corrosion in TPFPB-containing electrolyte, the characteristic two-stage voltage profiles can be well preserved within 50 cycles and the satisfactory

cycling stability is achieved. The discharge capacity remains at a high value of 200 mAh g⁻¹ after prolonged cycling over 100 cycles, corresponding to a minor capacity decay rate of 0.46%. Although the CE is slightly compromised in the early stage, its value shows a rising trend and approaches 95% with the progress of cycling. Under the high specific currents of 500 and 1000 mA g⁻¹, the reversible capacities are still preserved at 259 and 213 mAh g⁻¹ for the 3.0 V class cell (Fig. 3D), respectively, which are higher than those for the 2.7 V class cell (189 and 143 mAh g⁻¹, Supplementary Fig. 12B, C). In terms of specific energy and power (based on the mass of the active positive electrode materials, Fig. 3E and Supplementary Table 1), due to the low average discharge voltage (typically <1.0 V), oxides (e.g., TiO₂ and MnO₂) and sulfides (e.g., Cu₂S and VS₂) mostly show a specific energy below 300 Wh kg⁻¹ even under a low specific power around 100 W kg⁻¹. Despite the high operational voltage of insertion-type positive electrodes (e.g., LiFePO₄ and LiMn₂O₄), the limitation of available capacity (<150 mAh g⁻¹) caused by single-electron transfer reaction results in the specific energy below 200 Wh kg⁻¹ under the high specific power around 1000 W kg⁻¹. For conversion-type fluorides, the multi-electron redox center and highly electronegative F-ligand contribute to the combination of high specific capacity and high reaction voltage²¹. Consequently, under the specific currents of 100 and 1000 mA g⁻¹, our FeO_{0.7}F_{1.3} could offer the high specific energies of 485 and 267 Wh kg⁻¹ based on the discharge capacities of 354 and 213 mAh g⁻¹ with the average voltages of 1.37 and 1.26 V, corresponding to the specific powers of 137 and 1257 W kg⁻¹, respectively.

Raising the discharge cut-off voltage could reduce the multiphase transformation reactions involved, and it is beneficial for structure stabilization and cycling durability of fluoride hosts. We performed the charge-discharge measurements for intercalation-type Mg||FeO_{0.7}F_{1.3} cells using two APC-type electrolytes at 100 mA g⁻¹ in a voltage range of 1.1–2.5 V. As shown in Supplementary Fig. 13, the initial discharge profile displays the sole solid solution reaction stage without involving the conversion reaction, offering a dominant plateau at -1.75 V with a capacity beyond 200 mAh g⁻¹. As the cycling proceeds, the cell with LiCl-APC electrolyte suffers from the evident plateau degradation and capacity decrease. Note that the second-stage discharge plateau appears at below -1.5 V after 200 cycles (Supplementary Fig. 13A), indicating the occurrence of conversion reaction in fluoride positive electrode. The accumulation of converted metallic Fe is prone to induce anodic corrosion and parasitic side reactions, which should be responsible for the sharp CE decline in subsequent cycling (Fig. 3F). In comparison, the modified cell with TPFPB addition can well maintain the sloped intercalation plateau within 300 cycles without involving conversion reaction (Supplementary Fig. 13B). The accelerated desolvation process induced by TPFPB is expected to facilitate the interfacial charge transport, mitigate the positive electrode composition (or ion distribution) inhomogeneity and suppress the multiphase transformation in Li/Mg-concentrated domains, which is beneficial for the FeO_{0.7}F_{1.3} host to maintain structure stability during long-term cycling. When raising the charge cut-off voltages to 2.7 and 3.0 V, the reversible capacities of intercalation-type Mg||FeO_{0.7}F_{1.3} cell are boosted to 130 and 147 mAh g⁻¹, and the corresponding average discharge voltage is uplifted to 1.69 and 1.77 V, respectively (Fig. 3G). Although the voltage plateaus are slightly weakened during long-term cycling, probably attributed to the inevitable local structure degradation under increased ion flux, the FeO_{0.7}F_{1.3} still does not experience obvious conversion reactions (Supplementary Fig. 14). All the TPFPB-containing systems exhibit the good cycling stability up to 500 cycles (Fig. 3F), with the low average capacity decay rates of 0.056%, 0.043%, and 0.054% per cycle, and high average CE values of 98%, 97%, and 97%, for the 2.5, 2.7, and 3.0 V class cells, respectively. On the contrary, the baseline case shows a more rapid capacity attenuation, with a larger decay rate of 0.107%. At high specific currents of 500 and 1000 mAh g⁻¹, the reversible capacities are still preserved at 122 and

103 mAh g⁻¹ for the 3.0 V class cell (Supplementary Fig. 15), which are 25 and 32 mAh g⁻¹ higher than those for the 2.7 V class cell (Supplementary Fig. 16), respectively. After the fast electrochemical reaction, both the 3.0 V and 2.7 V class cells show high resilience with capacities of 155 and 125 mAh g⁻¹, respectively, as well as inspiring cycling stability when returning to a specific current of 100 mA g⁻¹. From the performance perspective, the initial capacity loss and CE (as shown in Supplementary Figs. 17 and 18) need to be further improved in future studies on the emerging Mg-fluoride batteries. Our study presents a promising research direction to develop electrolyte materials with a wide electrochemical window, enabling the good compatibilities with both high-potential fluoride positive electrodes and passivation-sensitive Mg negative electrodes.

We also investigated the impact of TPFPB additive concentration on the electrochemical performance of conversion-type Mg||FeO_{0.7}F_{1.3} cells (Supplementary Fig. 19). As expected, increasing the anion receptor addition can enhance its interaction with Cl-based ions and further alleviate the anodic corrosion/electrolyte decomposition, leading to the improved CE values approaching 97% for the cell using LiCl-APC-0.05 M TPFPB electrolyte. However, due to the relatively large molecular structure, the introduction of TPFPB would more or less increase the electrolyte viscosity and decrease bulk ionic mobility. As a result, the cell with 0.05 M TPFPB shows the slower reaction kinetics than that of the cell with 0.02 M TPFPB, with a lower reversible capacity and a larger overpotential of 850 mV (vs. 770 mV for the case of 0.02 M TPFPB). Therefore, an optimal amount (0.02 M) of TPFPB additive in LiCl-APC electrolyte enables an appropriate balance between interfacial stability and electrochemical kinetics.

Sub-zero Celsius performance has received less attention in Mg-based batteries, but it is a critical metric for assessing the practical applications of MMBs. The lower temperature environment would slow down the mass transport across the cell, and it poses a grand challenge to the emerging magnesium-fluoride conversion system in maintaining the high-capacity and high-energy release. Herein, the Mg||FeO_{0.7}F_{1.3} cell with LiCl-APC-TPFPB electrolyte is operated at -20 °C to investigate the low-temperature performance. This cell is pre-cycled at 25 °C for 5 cycles to activate the fluoride electrode via sufficient electrochemical grinding, and then it is transferred to the low-temperature box to conduct the galvanostatic discharge-charge testing at a specific current of 20 mA g⁻¹ in a voltage range of 0.5–3.0 V. As shown in Fig. 3H, the Mg||FeO_{0.7}F_{1.3} cell preserves a high capacity of 177 mAh g⁻¹ and good reversibility over 200 cycles at -20 °C. Note that lowering the temperature typically reduces the Mg-cation participation in positive electrode reaction due to the diffusion-limited reactivity, which is favorable for the reversible electrochemistry of dominated lithiation-driven phase transformation in fluoride. Meanwhile, the interfacial side reactions, such as Cl-corrosion and Fe-dissolution, are expected to be further mitigated in the low-temperature environment. These effects should be responsible for the improved cycling stability with a lower capacity decay rate of 0.18% per cycle for Mg||FeO_{0.7}F_{1.3} cell when the temperature is switched from 25 °C to -20 °C. The alleviated anodic corrosion also contributes to the high average CE value of 98%. In addition, the voltage polarization is well stabilized and the average discharge voltage is kept as high as 1.23 V (Supplementary Fig. 20). It can be attributed to the proven benefit of weakly solvating electrolyte with TPFPB-mediated solvation structure, which facilitates the interfacial ion desolvation and charge transfer behavior at low temperature^{36,37}. The above results demonstrate the satisfactory temperature tolerance of our proposed high-performance Mg||FeO_{0.7}F_{1.3} battery system.

The APC-type electrolytes feature the high ionic conductivity and good Mg-metal compatibility, providing a reliable medium for the shuttle of metal-ion charge carriers in Mg||FeO_{0.7}F_{1.3} cells. However, for LiCl-APC electrolyte, there are serious interfacial issues between the fluoride positive electrode and electrolyte under high-voltage

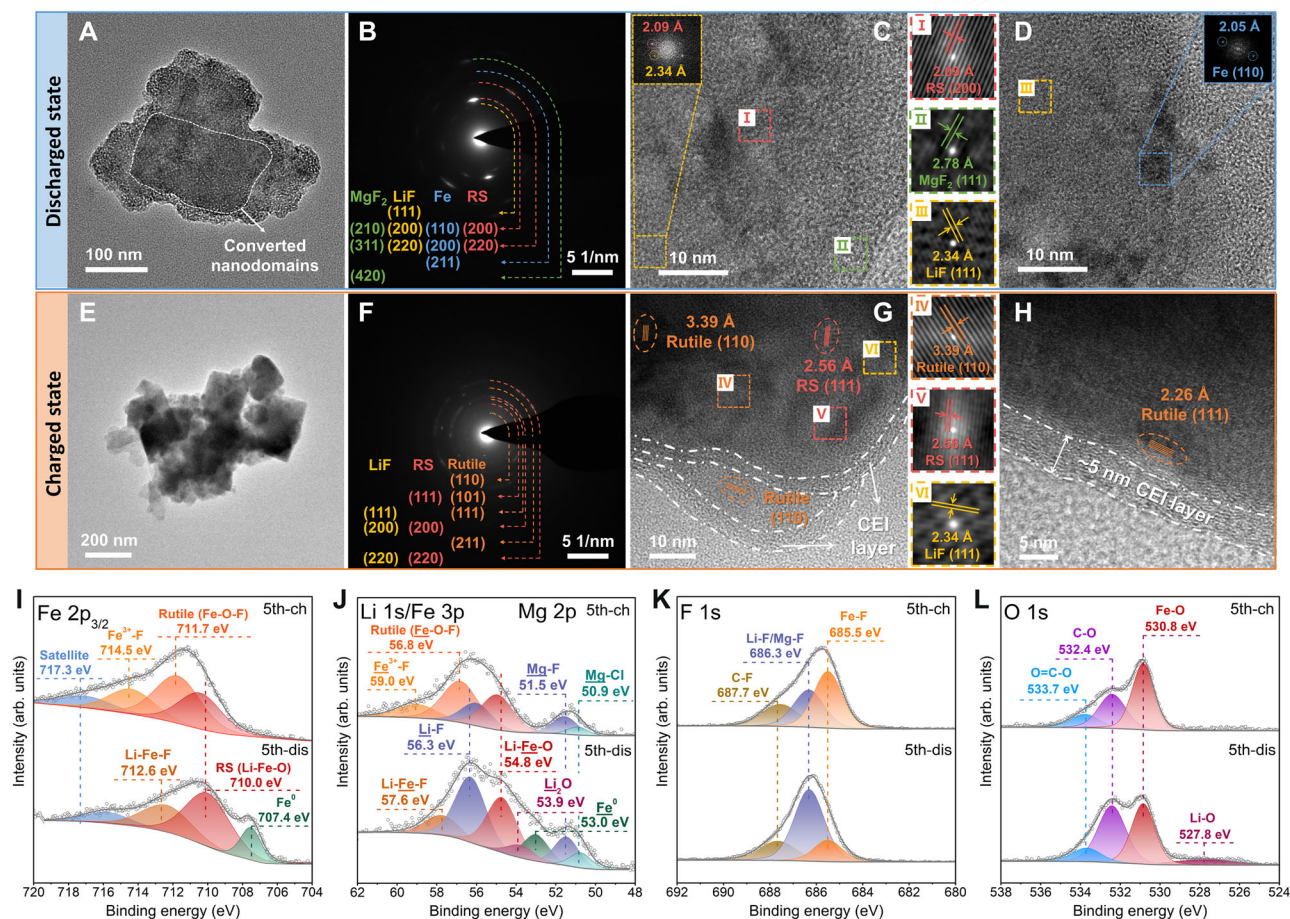


Fig. 4 | Bulk phase structure and surface chemical component evolutions of cycled $\text{FeO}_{0.7}\text{F}_{1.3}$ electrodes in LiCl-APC-TPFPB electrolyte. **A** TEM image, **B** SAED pattern, and **C**, **D** HRTEM images after discharging to 0.5 V, showing the dominant converted phases of rock salt (RS, Li_xFeO_2), metallic Fe, LiF and MgF_2 . **E** TEM image, **F** SAED pattern, and **G**, **H** HRTEM images after charging to 3.0 V, showing a conformal and homogenous cathode-electrolyte interphase (CEI) layer

with a thickness of ~5 nm covering particle surface. The insets sandwiched between HRTEM images display typical lattice stripes corresponding to discharged and charged products. XPS spectra of **I** Fe 2p_{3/2}, **J** Li 1s/Fe 3p/Mg 2p, **K** F 1s, and **L** O 1s for discharged (dis) and charged (ch) electrodes. Before ex situ characterizations, discharged and charged $\text{FeO}_{0.7}\text{F}_{1.3}$ electrodes were cycled at 100 mA g⁻¹ and 25 °C for 5 cycles.

operation. As illustrated in Fig. 3I, the chlorine-based electrolyte components, such as Mg_2Cl_3^+ would cause the oxidative corrosion to positive electrode material and its current collector. On the other hand, the divalent Mg^{2+} cation induces a strong solvation effect, thereby leading to the sluggish ionic desolvation (including dechlorination) and diffusion at electrode-electrolyte interface. Meanwhile, the Cl-based cations (e.g., MgCl^+) are prone to participate in the electrochemical reaction of positive electrode due to the high scission energy of metal-chloride bonds^{38,39}. Unfortunately, the electrode chlorination with transformation of transition-metal fluorides to chlorides (e.g., FeOCl and FeCl_2) would aggravate the dissolution of active species⁴⁰. These undesired interfacial activities give rise to the low CE and rapid capacity decay in baseline APC-type cells. Aiming at these issues, introducing an anion receptor of TPFPB in the electrolyte can firstly regulate the Lewis acid-base reactions of APC-type solution, and consequently reduce the major culprit of Mg_2Cl_3^+ for corrosion. Moreover, TPFPB exhibits the electrostatic interactions with Cl-based ions and THF solvent molecules. It not only alleviates the corrosion effects of Mg_2Cl_3^+ and Cl^- but also facilitates the desolvation/dechlorination of Li^+ and Mg^{2+} cations. Mitigating electrode chlorination is beneficial to the reaction-zone confinement of active transition-metal species. Therefore, the TPFPB-mediated chemistry enhances the interfacial stability and kinetics of APC-type electrolyte for unlocking the high-capacity, long-life, and wide-temperature $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries.

Reaction mechanisms for $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries

Different from the well-known electrochemical reactions of exclusively Li-driven iron oxyfluoride positive electrodes, it is unclear about the potential Li and Mg co-driven reaction mechanisms for the $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries. Transmission electron microscope (TEM) characterization was employed to disclose the structure evolution of cycled oxyfluoride positive electrodes in LiCl-APC-TPFPB electrolyte. When discharging the cell to 0.5 V, the electrochemically reduced $\text{FeO}_{0.7}\text{F}_{1.3}$ particles retain the original rectangle-like morphology (Supplementary Fig. 21) and consist of numerous dark nanodomains embedded in the light matrix (Fig. 4A). The high-resolution TEM (HRTEM) images (Figs. 4C, 4D and Supplementary Fig. 22) reveal that these dark nanodomains (less than 10 nm in size) are assigned to the Fe-containing phases of rock salt (RS, Li_xFeO_2) and metallic Fe, as indicated by the characteristic lattice fringes corresponding to the (200) plane for RS and (110) plane for Fe^{25,41}. The dominant LiF (111) plane with a *d*-spacing of 2.34 Å is identified in the surrounding region with light contrast. In addition, a few nanodomains belonging to MgF_2 phase are also occasionally found. The selected-area electron diffraction (SAED) pattern (Fig. 4B) further confirms the transformation from rutile-structured oxyfluoride to these cubic-structured phases. The conversion-type $\text{FeO}_{0.7}\text{F}_{1.3}$ typically undergoes the intercalation-extrusion reaction in the confined voltage region, generating the RS-phase Li_xFeO_2 , metallic Fe and F-containing components. The incompletely decomposed RS-phase has high crystallinity, in line with the

bright diffraction rings in SAED pattern and clear lattice fringes in HRTEM images. In contrast, LiF and MgF_2 formed by phase segregation are expected to have relatively weak crystallinity. The RS and other precipitated phases often present a compact contact manner, which should be responsible for the superposition of their reflections in the fast Fourier transform patterns of selected square areas (Fig. 4C and Supplementary Fig. 22B). Therefore, apart from the thermodynamically and kinetically favorable lithiation reactions, Mg-ions also participate in the F-extrusion process during electrochemical reduction of $\text{FeO}_{0.7}\text{F}_{1.3}$ electrode.

Upon the subsequent charging process, the RS-phase can serve as a parent phase to uptake the previously extruded Fe and F components driven by the expulsion of Li and Mg cations. Consequently, when the cell is recharged to 3.0 V, the active particles enable a reintegration of nanodomains with reduced intra-grain interfaces (Fig. 4E and Supplementary Fig. 23A). As expected, the rutile phase can be electrochemically synthesized according to the reappearance of characteristic (110) and (111) planes observed from the SAED pattern (Fig. 4F) and HRTEM images (Fig. 4G, H, Supplementary Figs. 23B, C). Besides, we occasionally identify the residual LiF phase in some regions, which can account for the partially incomplete reconversion from RS to the rutile phase. The retained RS structure exhibits the lattice expansion after solid-solution reaction with Fe and F incorporation, resulting in the transformation of the dominant plane from (200) to (111) with an enlarged *d*-spacing from 2.09 Å to 2.56 Å. Note that an amorphous coating with a thickness of ~5 nm is attached to the electrode particle surface after cycling. This cathode-electrolyte interphase (CEI) should stem from a combination of pristine ionic liquid capping and extrinsic electrolyte-derived species²⁵. The continuous and conformal CEI can serve as a protective layer to alleviate the undesired cathode-electrolyte interactions, such as Fe-species dissolution and Cl-associated corrosion. In addition, this surface layer probably serves as a filtration film to block the passage of Mg^{2+} ions with sluggish diffusion kinetics, thereby reducing the positive electrode reaction involving the less reversible magnesiation.

The more chemical state information about reaction mechanisms can be revealed by X-ray photoelectron spectra (XPS), as shown in Fig. 4I–L and Supplementary Fig. 24. For the discharged positive electrode, the peaks centered at 710.0 and 707.4 eV in Fe 2 $p_{3/2}$ spectra (Fig. 4I) indicate the formation of the dominant RS Li-Fe-O phase and zero-valent Fe phase^{42,43}. The Li-Fe-F peak found at 712.6 eV should originate from the minor FeF_3 phase in pristine sample that has not undergone the deep conversion reaction²⁵. These reduced Fe-containing species can also be identified in Fe 3 p spectra (Fig. 4J). Besides, the Li-F signal (at 56.3 eV, Li 1s) and Mg-F signal (at 51.5 eV, Mg 2 p) are detected in the same spectrum, correspondingly accompanied by the generation of a prominent peak at 686.3 eV in F 1s spectra (Fig. 4K)^{44,45}. This result confirms the Li/Mg co-driven conversion reaction involving the Fe-F bond breakage and F-component extrusion. A faint shoulder peak assigned to Li_2O emerges at 527.8 eV in O 1s spectra (Fig. 4L), suggesting the probable occurrence of partial O-component extrusion from Li-Fe-O phase after discharging to 0.5 V vs. Mg^{2+}/Mg ²⁵. This Li_2O signal is also found at 53.9 eV in Li 1s spectra. During the recharge process, Li/Mg extraction brings about the weakening or vanishing of signals associated with the precipitation phases (e.g., Fe, LiF, MgF_2 and Li_2O) and the simultaneous recovery of Fe^{3+} -F and rutile Fe-O-F signals (in F 1s, Fe 3 p and Fe 2 p). Note that the Fe 2 $p_{3/2}$ peaks shift to the positions with higher binding energies, indicating the appearance of high oxidation state species, including rutile Fe-O-F, RS-phase Li-Fe-O(-F), and defect trirutile FeF_3 . On the whole, the local bonding evolution of active materials in $\text{FeO}_{0.7}\text{F}_{1.3}$ electrode is consistent with the phase transformation disclosed by TEM results. According to Mg 1s and Cl 2 p spectra (Supplementary Fig. 24A, B), the Mg-Cl component shows a similar concentration evolution as other Li/Mg-species during cycling. But most of the Cl-

components are likely derived from residual salts on the outermost surface of the cycled positive electrode, and a small amount of MgCl^+ may participate in surface/subsurface redox reactions (as demonstrated later). The C 1s spectra related to CEI component can be deconvoluted into the C-C (284.8 eV), C-N (285.7 eV), C-O (286.7 eV), C=O (288.3 eV) and C-F (290.4 eV) moieties (Supplementary Fig. 24C)²⁵. The detected C-N signal stems from the imidazole group in ionic liquid BmimBF₄. It is indicative of a retained ionic liquid coating on pristine fluoride particles during sample preparation. In view of the intensified C-F signal after charging, the F-rich TPFPB seems to enable a fluorination-reinforced effect on CEI layer. The incorporation of highly covalent C-F bonds into CEI layer is beneficial for suppressing spontaneous electron transfer from electrolyte to positive electrode, thus reducing the undesired anodic activities at interface⁴⁶. For the cells using APC-type electrolytes, CEI should be a key component to improve interfacial stability that can prevent the Cl-related corrosion of electrolytes and deactivate the catalytic activity of positive electrodes via contact isolation and electronic insulation⁴⁷.

To obtain more chemical details about the spatial distribution of discharge products, XPS depth profiling was applied for the $\text{FeO}_{0.7}\text{F}_{1.3}$ positive electrode at discharged state using ion beam etching method (Supplementary Fig. 25). With surface elimination, the weakening of signals is observed for the organo-functional groups in CEI coating, while the peaks related to the underlying active species become more pronounced accordingly. From the concentration-depth profiles (Supplementary Fig. 26), the atomic fractions of Cl and Al elements show a descending trend with etching, which is different from those of the active elements such as Fe, F, Li, and Mg. Correspondingly, the Mg-F and Mg-Cl components show the opposite concentration evolutions from surface to subsurface, according to the increased peak area for Mg-F and decreased peak area for Mg-Cl in Mg 1s spectra (Supplementary Fig. 25D). It can be deduced that the detected Cl-based species are mainly residual chloride salts covering the positive electrode surface and are not deeply involved in fluoride reactions. The low atomic fraction (lower than 2%) of Cl verifies the role of anion receptor in mitigating the chlorination of positive electrode. No discernible B signal is detected in B1s spectra (Supplementary Fig. 25I), implying that the TPFPB additive can maintain its structural stability well during electrochemical cycling and thereby ensuring a lasting positive effect on interfacial stability and kinetics.

The first-principles calculations with DFT were conducted to gain the deeper insights into cation co-driven reaction mechanism based on diffusion kinetic analysis. Here, the climbing image nudged elastic band (CI-NEB) method was employed to simulate the potential diffusion pathways and calculate the corresponding energy barriers for Li and Mg ions in the bulk phase of stoichiometric FeOF. Given the kinetically favorable lithiation process, Li-ion diffusion behaviors are first studied. As shown in Supplementary Fig. 27, there are three anion channels (i.e., mixed O/F, F, and O channels) along [001] direction, which provide the potential pathways (denoted as path 1, 2, and 3, respectively) for cation migration in rutile-type FeOF structure⁴⁸. Along the three paths, the intercalated cations can pass through the metastable octahedral sites when hopping between adjacent stable octahedral sites (Fig. 5B), which requires activation energy to overcome the corresponding energy barrier. The calculated energy barriers of path 1, 2, and 3 for Li-ion migration are 1.39, 0.18, and 4.40 eV, respectively (Supplementary Fig. 27F). The highest energy barrier of path 3 should be attributed to the strong electrostatic interaction between intercalated cation and bivalent lattice O^{2-} with concentrated charge nature. The electrostatic force also drives the Fe atoms to move off-center towards the O atoms, thereby increasing the Fe^{3+} -Li⁺ repulsion in O channel due to shortened distance (Supplementary Fig. 27E). As a result, the octahedral sites for Li occupation are in off-center positions after lattice relaxation (Supplementary Fig. 27D). It can be deduced that the lithiation sites within the mixed and F channels (denoted as H1

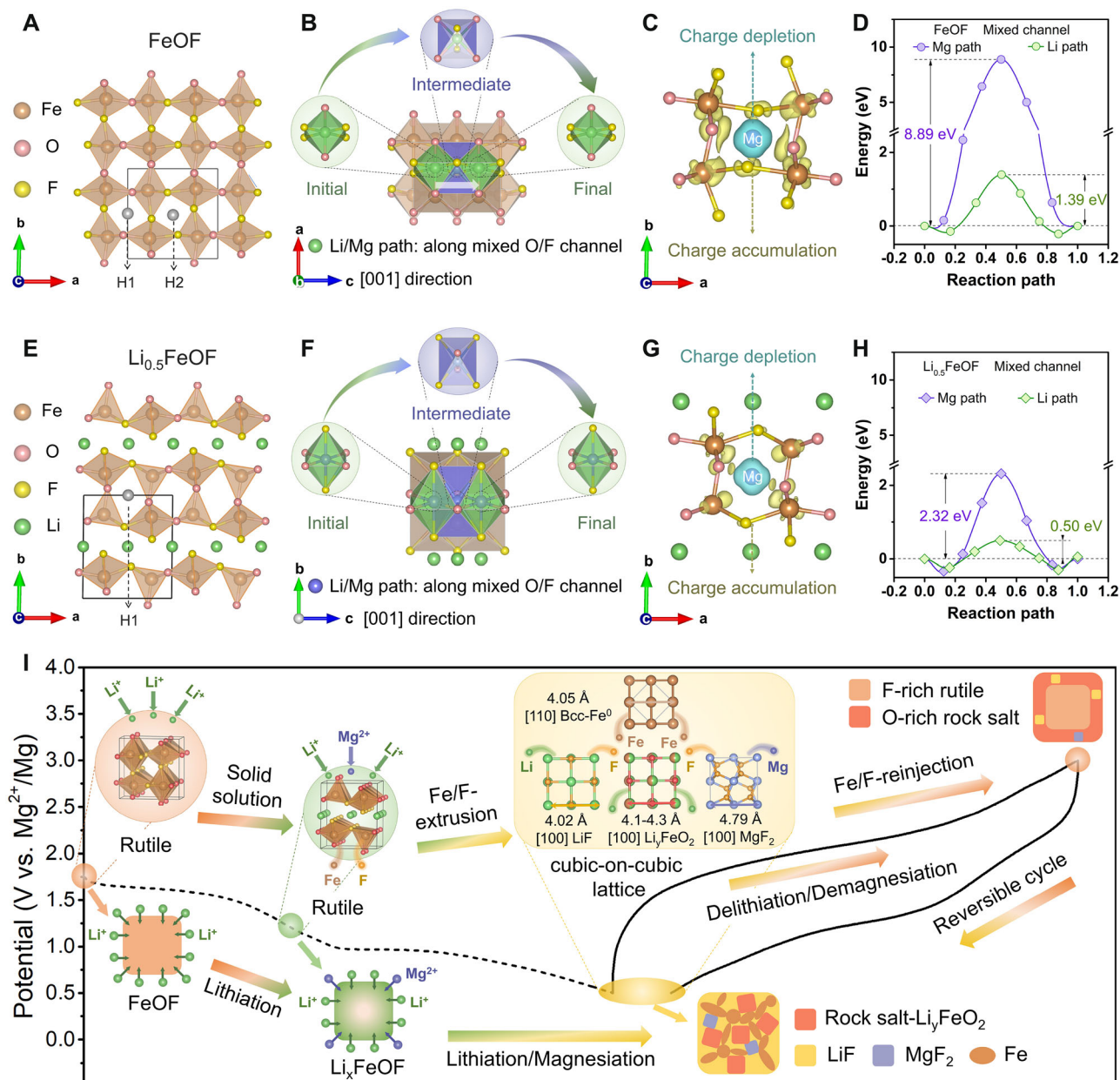


Fig. 5 | Solid-state diffusion behavior and reaction mechanism of rutile-type FeOF structure in $\text{Li}^+/\text{Mg}^{2+}$ -hybrid system. Favorable sites (H1/H2) for cation occupation in DFT-relaxed **A** FeOF and **E** $\text{Li}_{0.5}\text{FeOF}$ structures. Schemes of cation occupation of nearby stable octahedral sites in mixed O/F channels for **B** FeOF and **F** $\text{Li}_{0.5}\text{FeOF}$. Cation-hopping process between nearby stable octahedral sites (green color) would pass a metastable octahedral site (purple color). Illustration of

differential charge densities of **C** FeOF and **G** $\text{Li}_{0.5}\text{FeOF}$ with intercalated Mg atoms in mixed O/F channels. The yellow and blue isosurfaces represent charge accumulation and depletion regions, respectively. The isosurface level is set to $0.005 \text{ e}^-/\text{\AA}^3$. Calculated energy barrier profiles of Li-ion and Mg-ion diffusions along mixed O/F channels in the bulk of **D** FeOF and **H** $\text{Li}_{0.5}\text{FeOF}$. **I** Scheme of proposed redox mechanism and structure evolution of FeOF co-driven by Li and Mg cations.

and H2 respectively) are more energetically favorable or stable than those within the O channels (Fig. 5A). As for the Mg-ions with bivalent character, the migration energy barriers along the mixed (path 1) and F (path 2) channels are as high as 8.89 and 5.47 eV, respectively (Supplementary Fig. 28), indicating that the pristine rutile-structural FeOF likely blocks Mg^{2+} -ion diffusion and its participation in intercalation reaction. This should be an important reason for the good cycling stability of intercalation-type $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells.

When the content of intercalated Li reaches 0.5 to form $\text{Li}_{0.5}\text{FeOF}$, the separation between Fe and F atoms occurs along the favorable occupied F and mixed O/F channels in FeOF structure (Fig. 5E)⁴⁸. As shown in Supplementary Fig. 29, it causes the obvious lattice expansion along b-axis with the elongation of Fe-F and Fe-O bonds from

FeOF to $\text{Li}_{0.5}\text{FeOF}$. Notably, the volume of octahedral site for further cation occupation along the empty mixed O/F channel correspondingly increases by 29%. On the other hand, the Li-cation pre-intercalation also has an impact on the electron redistribution upon subsequent Mg insertion in $\text{Li}_{0.5}\text{FeOF}$. The differential charge density of Mg-ion intercalated in FeOF (Fig. 5C) shows a pronounced charge transfer from divalent Mg-ion to FeOF host, leading to the formation of charge depletion zone around Mg and charge accumulation region around adjacent O and F atoms. This explains why the migration kinetics of Mg-ions in FeOF structure is sluggish (Fig. 5D). As for $\text{Li}_{0.5}\text{FeOF}$, the charge accumulation on O/F ligands near Mg ion is markedly reduced (Fig. 5G), implying the weakened interaction between intercalated Mg-ion and $\text{Li}_{0.5}\text{FeOF}$ host. Overall, the expanded

channel spacing and weakened electrostatic interaction are expected to enhance the cation mobility along mixed O/F path (Fig. 5F), lowering the migration energy barriers to 0.50 eV for Li-ions and to 2.32 eV for Mg-ions (Fig. 5H). Such a significant reduction in diffusion barrier would encourage Li-ion together with its Mg-ion counterpart to co-drive the further bulk reaction in $\text{Li}_{0.5}\text{FeOF}$. Further cation insertion in $\text{Li}_{0.5}\text{FeOF}$ will increase the thermodynamic instability and induce the conversion reaction, leading to the preferential extrusion of fluoride components^{24,49}. It can be responsible for the identified MgF_2 in XPS and TEM results of the discharged oxyfluoride positive electrode.

Combining the above analysis of structural characterization and DFT calculation, we propose the reaction mechanism of iron oxyfluoride driven by dual-cationic charge carriers (Fig. 5I). The initial discharge process enables the solid-solution-like electrochemical lithiation from FeOF to Li_xFeOF ($0.5 \leq x < 1$), followed by the $\text{Li}^+/\text{Mg}^{2+}$ co-driven conversion reactions that result in the retained RS-phase Li_yFeO_2 and extruded Fe metal, LiF and MgF_2 (in the confined voltage region). The cubic lattice coherence and similar lattice constants of these discharge products tend to promote the local electronic and ionic transport at multiphase interfaces. It has a positive effect on the reinjection of Fe/F components into Li_yFeO_2 upon the delithiation/demagnesian (charge) process, thus being advantageous to reconstitute the parent structure and maintain the particle integrity. The reconverted F-rich rutile and O-rich RS phases, which are in tight contact at nanoscale (confirmed by TEM results), enable the good reversibility in voltage plateaus and capacity release during subsequent cycling for magnesium-fluoride batteries.

Kinetic behaviors of $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries

The conversion-type $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries undergo complicated structural and chemical changes in fluoride electrode, which will greatly impact the charge/mass transfer process and kinetic performance during electrochemical reaction. To study the kinetic characteristics of $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries, in-situ electrochemical impedance spectroscopy (EIS) was performed at different reaction stages throughout cycling process in a voltage range of 0.5–3.0 V. The corresponding potential-dependent Nyquist plots for a successive discharge-charge process are shown in Fig. 6A. Here we focus on the active reactions in fluoride electrode. Thus, the cell is pre-cycled for 10 cycles before EIS measurement to achieve the quasi-stable electrode–electrolyte interface and minimize the side-reaction interference. The Nyquist plots display a broad band from the high frequency of 100 kHz to the low frequency of 1.0 Hz, which contains the multiple resistance effects at electrode–electrolyte interface, including the electric double layer formation, ion transport through SEI/CEI layer ($R_{\text{SEI/CEI}}$), and charge transfer reaction (R_{CT})⁵⁰. The typical inclined straight line associated with the Warburg impedance of ion diffusion in porous electrodes is absent in the confined frequency region.

Considering the timescale fingerprint, the EIS spectra are quantitatively deconvolved for the distribution of relaxation time (DRT) analysis, thereby more clearly identifying the kinetic evolution of specific electrochemical processes⁵¹. As shown in Fig. 6B, the derived DRT plots at the beginning of discharge process show four distinct peaks within the time constant (τ) range of 10^{-7} – 10^1 s, denoted as τ_1 – τ_4 . The τ_1 peak at 10^{-6} s corresponds to the Z_{real} -axis intercept of Nyquist plots in the highest frequency region, which can be attributed to the bulk electrolyte resistance (R_b) and the electric resistance inside cell. With the progress of positive electrode lithiation, the concentration of kinetically favorable Li-ions in the hybrid electrolyte decreases due to the lack of replenishment from the Mg negative electrode. It likely causes the gradual growth of the τ_1 peak from 18 to 43 Ω with the decrease of discharge voltage. The low τ_2 peak at around 10^{-5} s is generally assigned to the electric double-layer relaxation, which is associated with the absorbed Li/Mg-ions at electrode–electrolyte interface⁵². Thus, the τ_2 peak shows a similar increasing trend as the τ_1

peak due to the gradually depletion of Li-cations at positive electrode side during discharge process. The impedance of mass transport across SEI and CEI typically occurs with a time constant of 10^{-4} to 10^{-2} s⁵¹. The slight variation in the τ_3 peak located at around 5×10^{-4} s indicates that the chemical composition of electrode–electrolyte interphase is relatively stable. The τ_4 peak related to charge transfer in cells reflects the interfacial electrode reactions. The rutile-type oxyfluorides typically experience the multi-stage intercalation-conversion reaction processes during discharge. Therefore, the corresponding time constants mainly vary in a large range of 10^{-2} to 1 s. The τ_4 peak first shows a decline in the early stage of intercalation reaction, which is in line with the viewpoint that a few Li-ions pillaring the host lattices can facilitate charge transfer⁵³. When the conversion reactions occur, the τ_4 peak splits into multiple peaks, implying that the multiphase transformation involves the complex reaction pathways, especially with the co-participating of Li and Mg ions. The overall charge transfer resistance shows an increasing trend due to the potential segregation and accumulation of insulating fluorides (LiF and MgF_2). Solid-state diffusion in the electrode is a sluggish process with a prolonged time constant usually more than 10 s, which is not involved in this study⁵¹.

Upon the subsequent charge, both the τ_1 and τ_2 peaks show an inverted decreasing trend with Li-extraction from positive electrode into electrolyte (Fig. 6C). It can be deduced that the charge transfer kinetics for re-conversion (charge) is poorer than that for conversion (discharge), as indicated by the larger resistance and longer timescale (slower relaxation rate) of corresponding τ_4 peak for re-conversion (charge). In addition, the τ_4 peak contributed by the phase reintegration process basically shows a single-peak evolution manner, which is different from the multi-peak evolution manner associated with the phase segregation process. This phenomenon suggests the asymmetric conversion paths for oxyfluoride during discharge and charge, as further verified by the observed asymmetric redox peaks in CV profiles (Supplementary Fig. 11). In the late charge, the τ_4 peak splits into a blurred doublet that likely corresponds to the partitioned reconstitution pathways of F-rich rutile and O-rich RS phases. Overall, the peak shape and peak value of DRT plots at the end of charge are comparable to those at the beginning of discharge, and after recharge τ_1 and τ_4 peaks recover to 23 and 608 Ω , close to their initial values of 18 and 422 Ω . This result well demonstrates the reaction reversibility of electrodes and good chemical stability of electrolyte and correlative interphases for $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries. According to the overview of DRT plot evolution in Fig. 6D, the resistance peak reaches the maximum in the response region of charge-transfer at the beginning stage of re-conversion (charge). It discloses that the electrochemical splitting of segregated fluorides (e.g., LiF and MgF_2) remains a kinetic bottleneck for metal-fluoride conversion systems.

To further elucidate the impact of TPFPB additive on fluoride reaction kinetics at cathode–electrolyte interface, we performed Tafel analysis to evaluate the charge transfer capability of $\text{FeO}_{0.7}\text{F}_{1.3}$ in Mg-based batteries. Given the narrow electrochemical stability window of baseline APC-type electrolyte, a consistent voltage range of 0.5–2.5 V is applied in $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ batteries for kinetic characterization. Figure 6E shows the typical Tafel behaviors of $\text{FeO}_{0.7}\text{F}_{1.3}$ electrodes in two APC-type systems. A key parameter of exchange current (i_0) obtained from Tafel analysis describes the faradaic activity at the equilibrium potential of 1.20 V⁵⁴, which is dominated by the conversion reaction in applied voltage region. The exchange current for $\text{FeO}_{0.7}\text{F}_{1.3}$ during cathodic process is measured based on Tafel equation as follows:

$$\eta = \frac{RT}{\alpha F} \ln i_0 - \frac{RT}{\alpha F} \ln i \quad (1)$$

where R and F are the gas constant and Faraday's constant, respectively, T denotes the absolute temperature (298.15 K), and α represents the charge transfer coefficient. When polarizing the system far enough

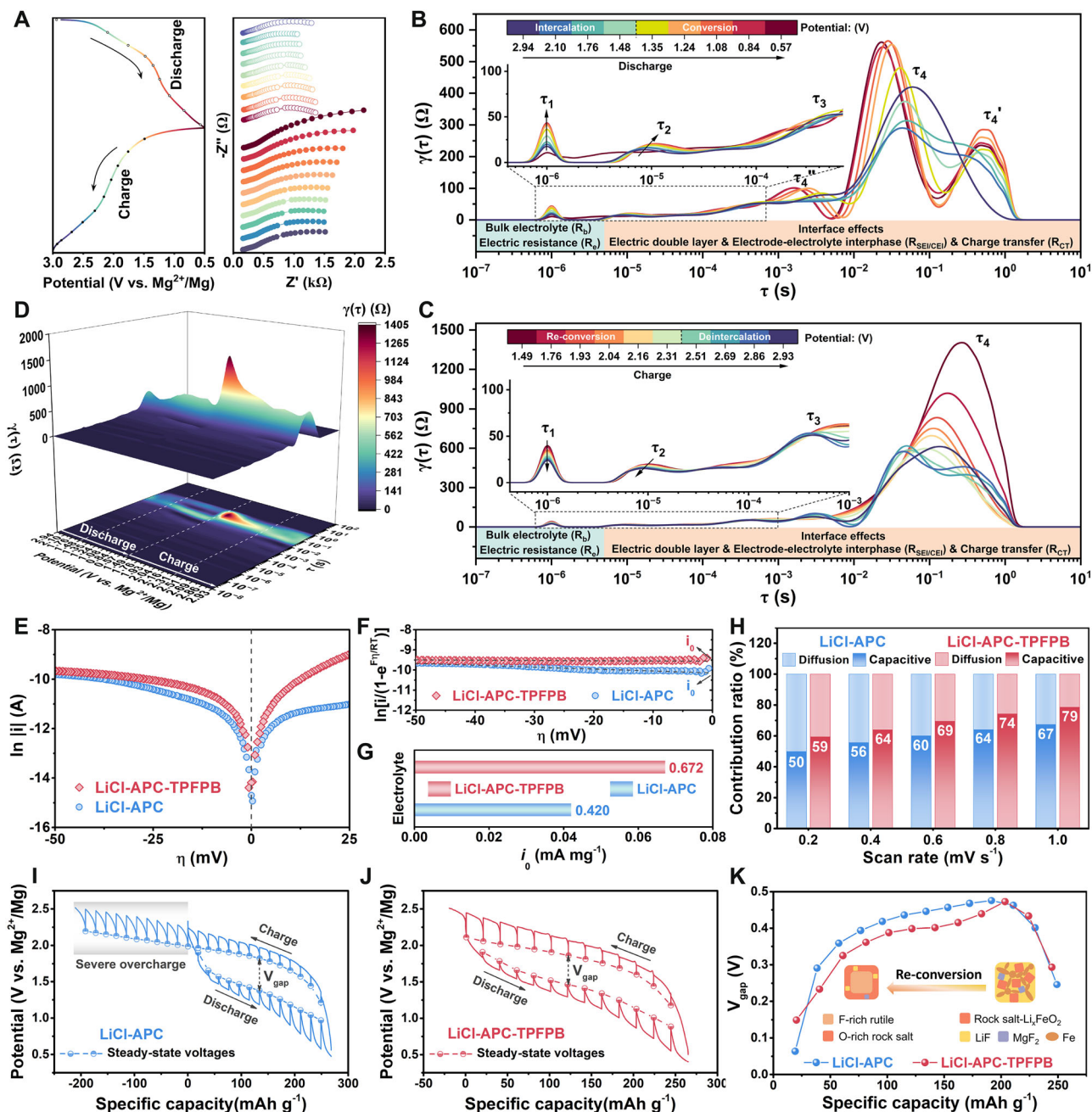


Fig. 6 | Reaction kinetics of $\text{FeO}_{0.7}\text{F}_{1.3}$ electrodes in APC-type electrolytes.

A Potential-dependent Nyquist plots of in-situ EIS during one full cycle and corresponding galvanostatic discharge and recharge curves. DRT plots obtained by deconvolving the EIS spectra during **B** discharge and **C** recharge processes. **D** DRT plot evolution during one full cycle. **E** Tafel plots of $\text{FeO}_{0.7}\text{F}_{1.3}$ electrodes in various electrolytes at the conversion plateau region during discharge process.

F Corresponding Allen-Hickling plots with linear fitting intercepts on vertical axis to determine exchange current. **G** Comparison of estimated exchange current in different electrolyte systems. **H** Normalized charge storage contributions from

capacitive (dark color) and diffusion-controlled (light color) reactions for $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells with different APC-type electrolytes in a voltage range of 0.5–2.5 V calculated from CV measurements. GITT curves of $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells using **I** LiCl-APC and **J** LiCl-APC-TPFPB electrolytes operated at 20 mA g^{-1} with an intermittent time of 1 h followed by an open-circuit relaxation for 6 h in a voltage range of 0.5–2.5 V. **K** Comparison of voltage gap (V_{gap}) between charge and discharge steps after relaxation at the same state of charge in the cases of different electrolytes. All electrochemical tests associated with reaction kinetics were performed at 25°C .

from the equilibrium state, the logarithm of current response (i) shows a linear relationship with the voltage polarization (η). As shown in Fig. 6F, the Tafel results on cathodic side can be replotted according to the Allen-Hickling plots based on the Butler-Volmer equation as follows:

$$\ln\left[i/\left(1 - e^{\frac{F\eta}{RT}}\right)\right] = \ln i_0 - \frac{\alpha F}{RT}\eta \quad (2)$$

Here, by linearly extrapolating the linear segment of Tafel plots and yielding an intercept ($\ln i_0$) on vertical axis, the exchange current at the equilibrium state can be estimated. Clearly, the TPFPB-containing system exhibits a higher i_0 value (0.672 mA mg^{-1}) than that of baseline system (0.420 mA mg^{-1}), confirming the faster charge transfer for the former (Fig. 6G). This kinetic advantage is ascribed to the favorable cation desolvation process induced by an anion receptor. The interface charge transfer behavior has an important impact on

surface or subsurface faradaic reaction of fluoride electrode. To give a quantitative analysis on the kinetic-dependent charge storage mechanism, CV measurements at various scan rates from 0.2 to 1.0 mV s⁻¹ were performed for Mg||FeO_{0.7}F_{1.3} cells with two APC-type electrolytes (Supplementary Figs. 30 and 31). During CV scanning, the faradaic current response (*i*) at a fixed potential (*V*) involves the contributions from capacitance effect (*k*₁*v*, related to the surface/subsurface redox reaction) and diffusion-controlled reaction (*k*₂*v*^{1/2}), according to the equation of $i(V) = k_1v + k_2v^{1/2}$, where *v* denotes the scan rate^{54,55}. This equation can be deduced as $i(V)/v^{1/2} = k_1v^{1/2} + k_2$, thereby the *k*₁ and *k*₂ constants are determined by the slope and intercept of the linear relationship between $i(V)/v^{1/2}$ and *v*^{1/2}, respectively. Supplementary Fig. 30B and 31D-F show the typical CV profiles at 0.6 mV s⁻¹, where the capacitive current is integrated into the dark region and distinguished from the total current. For FeO_{0.7}F_{1.3} in LiCl-APC-TPFPB electrolyte, the pseudocapacitive contribution is about 59–79% at various scan rates, evidently higher than that of 50–67% in the case of LiCl-APC (Fig. 6H). These results indicate that TPFPB additive accelerates the interfacial kinetics and facilitates the redox reactions at positive electrode (sub)surface. Note that the pseudocapacitive effect gradually weakens when the voltage range is widened from 0.5–2.5 V to 0.5–3.0 V (Supplementary Fig. 32). The anodic scanning to the higher voltage would motivate the segregated phases to reintegrate into the parent phase with an increase in grain size, which would reduce the nano-size effect and result in the higher-fraction diffusion-controlled process in bulk. Nonetheless, the favorable interfacial kinetics contributes to the well-preserved CV shape and well-defined redox peaks with the increase of scan rate (Supplementary Fig. 31A-C).

The galvanostatic intermittent titration technique (GITT) was performed to investigate the thermodynamic properties of cycled Mg||FeO_{0.7}F_{1.3} batteries, which are much dictated by the electrochemical kinetics at electrolyte–electrode interface. A series of GITT steps is applied in a voltage range of 0.5–2.5 V for the cells based on two APC-type electrolytes. Each GITT step comprises a short pulse with a constant specific current of 20 mA g⁻¹ for 1 h, followed by an open-circuit relaxation for 6 h. The long relaxation process enables the sufficient ion diffusion within fluoride electrode, allowing the chemical composition to become homogeneous⁵⁶. Figure 6I, J shows the GITT curves and obtained steady-state voltage profiles after relaxation (representing the quasi-thermodynamic equilibrium states, plotted with dashed lines). Compared with the TPFPB-containing system, the baseline LiCl-APC system suffers from a severe overcharge issue with large irreversible capacity generated at high voltage above 2.0 V. This phenomenon is attributed to the Cl-related anodic corrosion/electrolyte decomposition, which is aggravated by the eliminated reaction overpotential via relaxation process in GITT measurement. The fluoride conversion and reconversion involve multiphase redox reactions along with pronounced interfacial mass/charge transport phenomena. The intrinsic kinetic barriers typically cause the different spatial distributions (e.g., surface-to-core level) of electroactive phases for lithiated and delithiated fluoride electrodes⁵⁷. This compositional inhomogeneity is difficult to be completely eliminated via the normal relaxation process. As a result, the obtained steady-state reaction paths in GITT mode still deviate from the ideal thermodynamic equilibrium paths, leaving the voltage gap (*V*_{gap}) between charge and discharge steps at the same state of charge. Figure 6K displays the *V*_{gap} evolution of Mg||FeO_{0.7}F_{1.3} batteries using different APC-type electrolytes. At the initial stage of re-conversion process, there are the peak levels of *V*_{gap} values for both systems, corresponding to the deeply lithiated/magnesiated fluoride electrodes enriched with multiple heterogeneous phases. Then, the *V*_{gap} values show a downward trend as these discharge products of cubic phases reintegrate into the F-rich rutile and O-rich RS phases along with the reduction of heterogeneous interfaces upon recharging. Note that the *V*_{gap} values for LiCl-APC-TPFPB case are about 50 mV lower than those for LiCl-APC case in the

main conversion plateau region. The positive effect on the thermodynamic state of FeO_{0.7}F_{1.3} electrode during cycling should benefit from the improved interfacial stability and mass transport dynamics by TPFPB mediation on electrolyte composition and solvation structure.

Discussion

In summary, we developed a magnesium-fluoride battery prototype enabled by a TPFPB-mediated APC-type electrolyte. The boron-based receptor additive induces the interactions with Cl-based ionic species and THF solvent, thereby regulating the chemical composition and solvation structure of electrolyte. This anion-receptor chemistry reduces the magnesium chloride dimer cation responsible for anodic corrosion and significantly broadens the oxidation stability window of APC-type electrolyte. Additionally, the formation of loose solvation structures lowers the energy barriers for cation desolvation, especially the rate-determining dechlorination step. As a result, the tailored electrolyte simultaneously enhances the electrochemical stability and accelerates the charge transfer kinetics at electrode–electrolyte interface, endowing the Mg||FeO_{0.7}F_{1.3} batteries with high capacity, long-term cycling, and low-temperature tolerance. Based on the dual-cation co-driven conversion reactions, the proposed battery system can release the high reversible capacities of 354 mAh g⁻¹ at 100 mA g⁻¹ and 213 mAh g⁻¹ at 1000 mA g⁻¹. The corresponding specific energies reach 485 and 267 Wh kg⁻¹ under the specific powers of 137 and 1257 W kg⁻¹ (based on the mass of the active fluoride materials), respectively. Moreover, the Mg||FeO_{0.7}F_{1.3} batteries based on intercalation chemistry deliver a high average discharge voltage of 1.77 V and satisfactory durability over 500 cycles (with a low capacity decay rate of 0.054% per cycle). This work promises to flourish the research on exploring emerging magnesium-fluoride batteries for next-generation energy storage.

Methods

Synthesis of fluoride

The FeO_{0.7}F_{1.3} was prepared by a facile thermal treatment of hydrate iron fluoride, which was obtained based on a dissolution-precipitation method in an ionic liquid medium²⁵. Firstly, a certain amount of Ketjen Black (KB, Shenzhen Kejing Zhida Technology Co., Ltd., 25 wt% in the final weight of fluoride composite) was added into 10 mL of ionic liquid 1-butyl-3-methylimidazolium tetrafluoroborate (BmimBF₄, Aladdin, 98%), followed by vigorous stirring at 50 °C for several hours to form a uniform black solution. Then, it was transferred into the ice-water bath and cooled down to 0 °C, and 1.0 g of Fe(NO₃)₃·9H₂O (Aladdin, 98%) was gradually added into the KB-suspended solution under stirring state. The reaction solution was kept stirred for 12 h at the ice-water bath to allow sufficient fluorination. The precipitate product (FeF₃·3H₂O/KB) was washed with acetone (Adamas, 99.9%) and centrifuged several times, followed by drying under vacuum at 80 °C for 12 h. Finally, the hydrate iron fluoride composite was heated at 300 °C for 5 h with a ramping rate of 5 °C min⁻¹ to obtain the FeO_{0.7}F_{1.3} sample. The as-prepared FeO_{0.7}F_{1.3} sample contains a dominant phase of FeO_{0.77}F_{1.23} and a minor phase of FeF₃ at the molar percentages of 85.51% and 14.49%, respectively²⁵.

Electrolyte preparation

AlCl₃ (>98.0%) and tris(pentafluorophenyl)borane (TPFPB, >98.0%) were bought from Tokyo Chemical Industry Co. Ltd. Tetrahydrofuran (THF, ≥99.8%) was acquired from Alfa Aesar. Phenylmagnesium chloride (PhMgCl, ≥99.8%) and LiCl (≥99.99%) were purchased from Aladdin Reagent Co. The baseline all phenyl complex (APC)-type electrolyte was synthesized by reacting 0.25 M AlCl₃ and 0.5 M PhMgCl and adding 1.0 M LiCl in THF. Specifically, 0.1333 g of AlCl₃ was dissolved in 3 mL of THF with stirring overnight, and then 1 mL PhMgCl (2.0 M in THF) was added into the above solution. After the sufficient Lewis acid-base reactions, 0.1696 g LiCl was added into the solution

and continuously stirred until it was completely dissolved prior to use (denoted as LiCl-APC). The additive-containing electrolyte was obtained by dissolving 0.02 M or 0.05 M TFPFB into the above LiCl-APC solution. Among them, the optimal formulation with 0.02 M TFPFB is denoted as LiCl-APC-TFPFB and used for a comparative study with the baseline LiCl-APC. All the electrolytes were prepared inside an Ar-filled glove box with oxygen and moisture contents lower than 0.1 ppm at a room temperature of 25 °C.

Electrochemical measurement

The electrochemical performance was investigated by using CR2032 coin-type cells with the as-synthesized oxyfluoride as positive electrode and the high-purity magnesium foil (Hefei KE JING Materials Technology Co., LTD., 99.95%, 50 µm in thickness) as negative electrode. The positive electrode was prepared by homogeneously mixing the $\text{FeO}_{0.7}\text{F}_{1.3}$ powder, Super P carbon (Shenzhen Kejing Zhida Technology Co., Ltd.) and poly(vinyl difluoride) (PVDF, Sigma Aldrich, Mw = 534,000) binder with a weight ratio of 8:1:1 in the N-methyl-2-pyrrolidinone (NMP, Sinopharm Chemical Reagent Co., AR) solvent, followed by pasting the slurry on the 8-mm-diameter carbon paper (Toray, TGP-H-060, which was attached on a Mo current collector) and drying in a vacuum oven at 80 °C for 20 h to remove the solvent. The prepared positive electrode contains the active fluoride material with a mass loading of about 1.0 mg cm⁻² and has a thickness of about 24 µm. The negative-to-positive electrode capacity ratio (N/P ratio) is estimated to be about 63.6. A glass microfiber membrane (Whatman GF/B, cat. No. 1821-090, thickness = 0.68 mm, average pore size = 1.0 µm) with a diameter of 19 mm was employed as the separator. The APC-type solutions with and without TFPFB additive were used as the electrolytes. The volume of electrolyte injected in each coin cell is controlled at about 90 µl by pipette. The whole cell assembly process was carried out in an Ar-filled glovebox.

Regular galvanostatic charge-discharge tests were operated at room temperature (25 °C, if not specially indicated) on a LAND-CT2001A battery system at various specific currents from 100 to 1000 mA g⁻¹. The charge cut-off potential is set at 2.5, 2.7, or 3.0 V vs. Mg^{2+}/Mg . The discharge cut-off potential is set at 1.1 V for intercalation reaction or at 0.5 V for additional conversion reaction. The cell for low temperature test was pre-cycled at 25 °C for 5 cycles and then transferred to an environment of -20 °C to conduct the following electrochemical cycling at a specific current of 20 mA g⁻¹ in a voltage range of 0.5–3.0 V. GITT was operated at 25 °C and a low specific current of 20 mA g⁻¹ for 1 h followed by open-circuit relaxation for 6 h to reach a quasi-equilibrium state. Cyclic voltammetry (CV), linear sweep voltammetry (LSV), Tafel analysis, and in-situ electrochemical impedance spectroscopy (EIS) were performed on an electrochemical workstation (VersaSTAT3, AMETEK Scientific Instruments) at 25 °C. CV measurements of $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cells were run at different scan rates from 0.1 to 1.0 mV s⁻¹ in various voltage ranges of 0.5–2.5 V, 0.5–2.7 V, and 0.5–3.0 V. To evaluate the oxidation stability of APC-type electrolytes and their anodic corrosion on metallic current collectors, LSV tests were conducted at a scan rate of 0.1 mV s⁻¹ using a two-electrode or three-electrode configuration. Stainless steel (SS) or Mo foil was used as the working electrode, and Mg foils serve as the counter electrode for two-electrode configuration and as both the reference electrode and counter electrode for three-electrode configuration, respectively. The reversibility of Mg plating/stripping process in APC-type electrolytes was evaluated by conducting the galvanostatic cycling in $\text{Mg}||\text{Cu}$ asymmetric cells at a current density of 0.5 mA cm⁻² and an areal capacity of 1.0 mAh cm⁻². The Cu foil (99.9%) with a thickness of 12 µm was acquired from Hefei KE JING Materials Technology Co., LTD. Tafel analysis was performed on the 10th discharge process by holding the voltage at the conversion plateau of 1.20 V for 1 h, followed by a linear scan at 1.0 mV s⁻¹ within the voltage range of ±100 mV around the

open-circuit voltage. For in-situ EIS measurements, the $\text{Mg}||\text{FeO}_{0.7}\text{F}_{1.3}$ cell was pre-cycled for 10 cycles at 100 mA g⁻¹ in a voltage range of 0.5–3.0 V to allow the electrodes and related interfaces to reach a quasi-stable state. Then, the cell intermittently proceeded with a galvanostatic discharge-charge cycle at 100 mA g⁻¹ in a voltage range of 0.5–3.0 V, with a specific capacity interval of 35 mAh g⁻¹, followed by an open-circuit operation for 30 min and a potentiostatic EIS measurement to record the Nyquist plots. Each potentiostatic EIS measurement was conducted under the open-circuit condition with an applied amplitude of 10 mV, in a frequency range from 100 kHz to 1.0 Hz (12 points per decade). The obtained Nyquist plots are deconvolved for the DRT analysis using pyDRTtools software⁵⁸.

Material characterization

Nuclear magnetic resonance (NMR) spectroscopy and Raman spectroscopy were performed to identify the effect of TFPFB mediation on the composition and solvation structure of APC-type electrolytes. The ¹¹B and ⁷Li NMR spectra were collected on a Bruker Avance III HD 400 MHz spectrometer. Raman spectra were measured with a Renishaw in Via Reflex confocal Raman microscope using a 633 nm laser. The ionic conductivities of baseline APC-type electrolyte and TFPFB-containing electrolyte were measured by a DDSJ-308A conductivity meter (Leici) at 25 °C. Scanning electron microscopy (SEM, Magellan 400L, FEI) was employed to observe the morphology of pristine $\text{FeO}_{0.7}\text{F}_{1.3}$ grains. Transmission electron microscopy (TEM, JEOL JSM-6700F, operated at 200 kV) was carried out to collect the (high-resolution) TEM images and selected area electron diffraction (SAED) patterns, which are used to disclose the morphology and microstructure of oxyfluoride positive electrodes at terminal discharged and charged states. The (sub)surface composition and bonding evolutions of cycled $\text{FeO}_{0.7}\text{F}_{1.3}$ electrodes were analyzed by X-ray photoelectron spectroscopy (XPS, ESCALAB-250) with an Al anode source. For the ex-situ TEM and XPS characterizations, the discharged or charged $\text{FeO}_{0.7}\text{F}_{1.3}$ electrode was cycled at 100 mA g⁻¹ and 25 °C for 5 cycles, and then taken out from the cells in an Ar-filled glove box. Subsequently, the collected electrodes were repeatedly washed by THF solvent to ensure the removal of residual salts. They were then dried overnight in an argon atmosphere before further characterization.

Computational details

Quantum chemical calculations were carried out using the Gaussian 09 (Revision D.01)⁵⁹ software package based on density functional theory (DFT) method. The geometry optimization and single-point energy calculation were performed at the B3LYP⁶⁰/6-311 + G(d,p)⁶¹/MO62X⁶²/def2TZVP⁶³ level of theory. The electrostatic potential (ESP) mapping of molecular surface with ESP surface extrema was obtained using Multiwfn (version 3.8) software⁶⁴ and Visual Molecular Dynamics (VMD, version 1.9.4a53) program⁶⁵ based on the above single-point energy calculation. Supplementary Data 1–6 provide the optimized configurations of THF solvent and Cl⁻ and AlCl_4^- anions individually, as well as their respective optimized configurations when interacting with TFPFB. The desolvation energy profiles of solvated clusters for Mg^{2+} -ion and Li⁺-ion were calculated based on the paths involving sequential solvent detachment and anion dissociation. The energy barrier ($E_{\text{desol.}}$) of stepwise desolvation was calculated by the following equation:

$$E_{\text{desol.}} = E_{\text{solvent/ion}} + E_{\text{res.}} - E_{\text{tot.}} \quad (3)$$

where $E_{\text{solvent/ion}}$, $E_{\text{res.}}$, and $E_{\text{tot.}}$ represent the energies of stripped species (THF solvent, Cl⁻, or [TFPFB-Cl]⁻), residual stable configuration after stripping, and initial stable configuration before desolvation step, respectively. Supplementary Data 7–10 provide the initial and final configurations of the solvated clusters for the hexa-coordinated

$\text{Mg}^{2+}(\text{THF})_5\text{Cl}^-$ and tetra-coordinated $\text{Li}^+(\text{THF})_3\text{Cl}^-$ with and without TFPFB mediation during structure optimization.

The structural relaxation of FeOF and $\text{Li}_{0.5}\text{FeOF}$ was performed by DFT calculation using the Vienna Ab initio Simulation Package (VASP, version 5.3.3) with Perdew–Burke–Ernzerhof parametrization of the generalized gradient approximation (GGA). The projector-augmented wave (PAW) pseudopotentials (version 5.2) were implemented in VASP to describe the electron-ion interaction. The PAW pseudopotentials and valence configurations for each atom used in DFT calculations are: Mg_pv ($2p^6 3s^2$) for magnesium, Li_sv ($1s^2 2s^1$) for lithium, Fe_pv ($3p^6 3d^7 4s^1$) for iron, O ($2s^2 2p^4$) for oxygen, and F ($2s^2 2p^5$) for fluorine. Considering the strong correlation character of Fe 3d electrons, a Hubbard-type correction (GGA + U) with a U_{eff} value of 5.0 eV was applied⁶⁶. A gamma-centered k-points mesh of $5 \times 5 \times 5$ was used to sample the Brillouin zone for both FeOF and $\text{Li}_{0.5}\text{FeOF}$. The activation barrier and migration pathway for charge-carrier (Mg^{2+} or Li^+) hopping were estimated by the climbing image nudged elastic band (CI-NEB) method⁶⁷. The potential migration pathways were constructed using the linear interpolation between initial and final states, and the structures were then relaxed with a maximum residual force threshold of $0.05 \text{ eV } \text{\AA}^{-1}$ for each atom. The cutoff energy was set to 450 eV, and the energy convergence criteria for structure relaxation were less than 10^{-6} eV . The force convergence criterion was set as $0.02 \text{ eV } \text{\AA}^{-1}$ for the structural relaxation of initial FeOF and $\text{Li}_{0.5}\text{FeOF}$. The energy convergence criterion was less than 10^{-6} eV for both structure relaxations and CI-NEB calculations. The cutoff energy was set to 450 eV in all the calculations. The crystal structures of FeOF and $\text{Li}_{0.5}\text{FeOF}$ are obtained from Material Project⁶⁸ and re-constructed according to the previous study⁴⁸ using VESTA software (version 3.4.0)⁶⁹. The crystal structures, clusters and differential charge density maps are visualized using VESTA software⁶⁹. The schematic illustrations of the regulating effect of TFPFB on electrolyte and the reaction mechanism of FeOF co-driven by Li/Mg-ions, as well as the format backgrounds of electrochemical performance diagrams, are generated using PowerPoint software. Supplementary Data 11, 12 provide the optimized structures of FeOF and $\text{Li}_{0.5}\text{FeOF}$ with Mg-ion intercalation.

Data availability

The data supporting the findings of this work are available within the article and its Supplementary Information file. The atomic coordinates of the optimized computational structures are provided as plain text files in Supplementary Data 1–12, packaged in a single zip file. Source Data are provided with this paper. The raw datasets that support the findings of this study are available from the corresponding author upon request. Source data are provided with this paper.

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Author contributions

C.L. proposed the idea and designed the project. K.C. developed the methodology and performed the experiments with the help of T.W. and G.L. M.L. and K.C. conducted the theoretical calculations. K.C. and C.L. analyzed the results and wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Chilin Li.

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